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RECEIVED 25 February 2026

REVISED 22 April 2026

ACCEPTED 23 April 2026

PUBLISHED 20 May 2026

CITATION

Rodriguez-Mata AE,
Gonzalez-Huitron VA, Flores-Colunga G,
Lozoya-Ponce R,
Acosta Cano de los Rios P and
Beltran-Gomez O (2026) Output
fractional-order sliding modes for
structural energy shaping: application to
robust brushless DC (BLDC)
traction systems.
Front. Control Eng. 7:1817112.
doi: 10.3389/fcteg.2026.1817112

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Output fractional-order sliding modes for structural energy shaping: application to robust brushless DC (BLDC) traction systems

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This paper presents an observer-based fractional-order sliding-mode control framework for nonlinear Brushless DC (BLDC) traction systems, combining a filtered high-gain observer (FHGO) with a fractional-order sliding-mode controller (FOSMC). The fractional exponent β is interpreted not merely as a tuning parameter, but as a structural energy-shaping and bifurcation variable that continuously modifies the dissipative geometry of the sliding manifold, generating a transition from integral-type regulation at $\beta = 1$ to nonlinear fractional dissipation for $0 < \beta < 1$. A Lyapunov-based analysis is developed to establish exponential convergence under nominal conditions and input-to-state stability in the presence of bounded load disturbances, measurement noise, and approximation residuals associated with the finite-order realization of the fractional operator. Performance is evaluated through both classical tracking indices, namely, RMSE, IAE, and ITAE, and structural energy metrics, including control energy J_u , control smoothness $J_{\dot{u}}$, control deviation energy J_{e_u} , time-weighted control deviation $ITAE_u$, and the sliding-surface deformation index J_s . Numerical simulations on a nonlinear dq Brushless DC (BLDC) traction model show that fractionalization reshapes the closed-loop energy distribution, significantly reducing accumulated tracking error while maintaining bounded actuation effort. The results reveal that an intermediate fractional regime, particularly within $\beta \in [0.6, 0.9]$, provides the most favorable compromise between robustness, smoothness, and structural efficiency. In addition, the proposed FHGO-filtered-surface-fractional-control interconnection leads to a clear attenuation of chattering, although not to its complete elimination. These findings show that fractional sliding control should be understood not as a simple

parametric extension of classical sliding-mode control, but as a geometric energy-shaping mechanism for structurally adaptive nonlinear regulation in electric vehicle drives.

KEYWORDS
brushless DC (BLDC) motor, electric vehicle drives, filtered high-gain observer, fractional-order sliding mode control, input-to-state stability, robust control

1 Introduction

Traditional control methodologies, such as Direct Torque Control (DTC) and Field-Oriented Control (FOC), have been extensively employed to address performance issues in electric drive systems. Although effective under specific steady-state operating conditions, these strategies often exhibit reduced performance under rapid load variations, strong dynamic transients, and severe operating scenarios [Chen \(2025\)](#), [Kamalapur and Aspalli \(2023\)](#). In response to these limitations, recent research has increasingly shifted toward advanced control paradigms that integrate applied mathematical tools with modern automatic control theory, including sliding mode control (SMC), fractional-order control, and predictive control schemes. These approaches aim to improve robustness, closed-loop stability, and transient performance in the presence of uncertainties, disturbances, and nonlinear effects [Ben Salem et al. \(2024b\)](#), [Jain \(2023\)](#), [Prabhu et al. \(2023\)](#).

Among these methodologies, fractional-order sliding mode and hybrid direct torque control schemes have shown significant effectiveness. The integration of adaptive fractional-order sliding mode control (SMC) with direct torque control-support vector machine (DTC-SVM) has demonstrated improvements in efficiency and robustness within electric vehicle propulsion systems [Ben Salem et al. \(2024a\)](#). In contrast, the application of fractional-order SMC in induction motor drives provides a swift

dynamic response while reducing torque ripple [Gudey et al. \(2023\)](#). Similarly, the application of sliding mode control in switching reluctance motors has shown effectiveness in reducing torque ripple, surpassing the performance of conventional PI controllers [Vinh and Ha \(2023\)](#). Enhanced direct torque control methodologies, incorporating composite controllers or fractional-order approaches, have been proposed to improve disturbance rejection and tracking accuracy in permanent magnet synchronous motor drives [Le and Hsieh \(2024\)](#), [Nicola et al. \(2024\)](#). Moreover, comparative evaluations of fractional high-order sliding mode control techniques highlight their robustness and reduced chattering, thereby improving their suitability for electric vehicles and renewable energy systems [Almakki et al. \(2023\)](#).

These advancements confirm a developing trend: modern electric vehicle motor control relies on complex automatic control techniques—rooted in sophisticated mathematics and applied sciences—to overcome the limitations of conventional methods, ensuring reliable operation, reduced torque ripple, and enhanced propulsion performance.

The fractional-order sliding mode control (FOSMC) paradigm has surfaced as a noteworthy methodology in the realm of advanced control theory, primarily due to its ability to proficiently handle nonlinear dynamics and improve resilience against external disturbances and parametric uncertainties. Unlike conventional SMC, fractional-order formulations integrate memory and smoothness traits that enhance system performance. [Ben Salem](#)

TABLE 1 Definition and interpretation of performance indices.

Index	Mathematical definition	Physical/Control interpretation
RMSE	$\sqrt{\frac{1}{T} \int_0^T e_q^2(t) dt}$	Measures the global tracking accuracy. It penalizes large deviations and reflects overall regulation quality across the entire trajectory
IAE	$\int_0^T e_q(t) dt$	Quantifies the accumulated absolute tracking error. Sensitive to persistent deviations and useful to evaluate total regulation performance
ITAE	$\int_0^T t e_q(t) dt$	Penalizes late-time errors more heavily. Indicates how quickly the controller suppresses deviations and achieves steady-state convergence
J_u	$\int_0^T u^2(t) dt$	Represents the total control energy. It reflects the actuation effort required by the controller and is directly related to power consumption and actuator demand
$J_{\dot{u}}$	$\int_0^T \dot{u}^2(t) dt$	Measures control smoothness and high-frequency activity. It is strongly associated with chattering, actuator stress, and mechanical wear
J_{ϵ_u}	$\int_0^T (u_{\beta=1}(t) - u_{\beta}(t))^2 dt$	Quantifies the structural deviation of the fractional controller relative to the integer-order References case. Indicates how much the control law is altered by fractionalization
$ITAE_u$	$\int_0^T t u_{\beta=1}(t) - u_{\beta}(t) dt$	Evaluates persistent differences in control action over time. Highlights long-term structural discrepancies introduced by fractional dynamics
J_s	$\int_0^T (s_{\beta=1}(t) - s_{\beta}(t))^2 dt$	Measures deformation of the sliding manifold induced by fractionalization. Captures geometric changes in the closed-loop dynamics and dissipative structure

(2024b) demonstrate that the integration of fractional-order sliding techniques with direct torque control and space vector modulation (DTC-SVM) significantly reduces torque ripple and flux oscillations in induction motor drives. Cheng (2025) demonstrated the improved effectiveness of fractional-order sliding mode observers in the sensorless control of permanent magnet synchronous motors (Brushless DC (BLDC)s), effectively addressing chattering issues and enhancing control accuracy. Recent studies highlight that improved direct torque control formulations employing fractional fuzzy logic controllers and adaptive sliding mode control result in notable enhancements in robustness and stability for electric vehicle propulsion systems Kamalapur and Aspalì (2023), Ben Salem et al. (2024a), Gudey et al. (2023).

High-gain observers (HGOs), particularly in their fractional and filtered variations, have demonstrated significant efficacy in delivering accurate state estimates in the presence of measurement noise and varying observability conditions Safder (2024), Liu et al. (2025). Recent advancements include fractional-order terminal sliding mode observers, which achieve improved accuracy in forecasting position and velocity—crucial parameters for sensorless motor operations Cheng (2025), Shen et al. (2025). Adaptive observer-based methodologies improve robustness and reduce computational complexity, enabling practical application in real-time electric vehicle drives Saha and Singh (2025), Khan et al. (2025). The advancements converge into a synergy between observer and controller, aligning with the theoretical foundations of applied mathematics, while addressing tangible practical challenges in electric vehicle propulsion.

This study introduces an extensive control framework that integrates fractional-order sliding mode controllers alongside high-gain observers specifically tailored for brushless DC (Brushless DC (BLDC)) motors utilized in electric vehicle applications, drawing on the latest developments in the field. The impetus arises from the necessity to enhance motor efficiency, minimize torque ripple, and ensure robustness against parametric variations and external disturbances. The proposed approach entails the development of fractional-order sliding mode controllers, augmented by filtered high-gain observers, validated through extensive numerical simulations and experimental configurations. This dual-layer architecture utilizes the strengths of sliding modes, the smooth convergence properties of fractional calculus, and the accuracy of advanced observer techniques, offering a solution that is both practical and grounded in theory. In other hands, Recent advances in nonlinear control have also explored dynamic surface control (DSC) with novel nonlinear filtering strategies that eliminate repeated differentiation and ensure global boundedness of closed-loop signals, even under parameter uncertainty Liu et al. (2025). These developments highlight the growing role of filtering mechanisms in robust control design. In this context, the proposed FHGO–fractional sliding-mode framework shares a similar philosophy by incorporating filtering dynamics; however, it differs fundamentally in that the fractional exponent β introduces a structural energy-shaping mechanism that modifies the dissipative geometry of the closed-loop system.

The outcomes of this initiative are anticipated to significantly improve electromobility technology, particularly in developing economies where infrastructural and financial barriers persist Salgado-Conrado et al. (2024), Villa-Mendoza et al. (2023),

Vallarta-Serrano et al. (2022). The proposed framework aims not only to optimize technology but also to improve industrial competitiveness and promote global environmental objectives by reducing emissions and enhancing the energy efficiency of transportation systems. This research combines scholarly inquiry, practical technological progress, and the principles of control theory to address contemporary engineering challenges, while also establishing a foundation for future enhancements in the advanced control of electric vehicles. Vinh and Ha (2023), Nicola et al. (2024), Almakki et al. (2023), Prabhu et al. (2023). The performance indices used throughout this study are defined and interpreted in Table 1.

1.1 Contribution of the article

This study offers a structural energy-based examination of fractional-order sliding mode control as applied to a nonlinear Brushless DC (BLDC) traction model that serves as a representative of electric vehicle drives. In contrast to previous research, the fractional effects are analyzed using both traditional tracking indices and manifold deformation metrics, thereby establishing a cohesive framework for performance and structural assessment. The contributions are as follows: (i) the design of a fractional sliding mode controller that is integrated with a filtered high-gain observer, (ii) the introduction of a multi-metric energy evaluation methodology for fractional tuning, and (iii) the identification of the exponent β as a bifurcation parameter that facilitates a continuous transition between PI-type regulation ($\beta = 1$) and nonlinear high-order sliding dynamics ($0 < \beta < 1$). The findings offer valuable insights for choosing the most suitable fractional regime and pave the way for the development of structurally adaptive controllers that can shift between linear and fractional sliding strategies.

It should be clarified that the novelty of this work is not claimed in the mere combination of a fractional-order sliding-mode controller with a filtered high-gain observer, since both ingredients have already been explored separately in the motor-control literature. Rather, the contribution lies in the structural interpretation of the fractional exponent β as an energy-shaping and bifurcation parameter within the proposed observer-based Brushless DC (BLDC) traction architecture. Under this viewpoint, β is not treated as a conventional gain-like tuning knob, but as a variable that continuously deforms the dissipative geometry of the filtered sliding manifold, thereby generating a transition between the integer-order reference configuration ($\beta = 1$) and the fractional nonlinear regimes ($0 < \beta < 1$). This distinction is further supported by the introduction of structural energy metrics that quantify not only tracking performance, but also control smoothness, energetic deviation, and manifold deformation.

2 Mathematical modeling and problem statement

Permanent Magnet Synchronous Motors (Brushless DC (BLDC)s) have become the favored choice for traction in electric vehicles, due to their compact dimensions, high efficiency, excellent power density, and advantageous torque response Mali and Susitra

(2024); Cherif et al. (2024); Ankit (2024). The nonlinear and coupled dynamics of Brushless DC (BLDC)s present significant challenges for advanced control design, particularly because of electromechanical coupling and sensitivity to parameter variations Ye et al. (2023); Kasri et al. (2024). A precise mathematical model is essential for capturing the underlying physics of the machine and establishing the analytical foundation necessary for robust control design, stability analysis, and real-time implementation.

2.1 Electrical subsystem: park transformation

The equations for the Brushless DC (BLDC) stator are most effectively articulated within the synchronously rotating dq reference frame, known as the Park transformation. This approach aligns the d -axis with the rotor flux, thereby decoupling the dynamics into direct and quadrature components. For a generic variable $\zeta \in \{i, v, \lambda\}$, the transformation from three-phase coordinates (abc) to ($dq0$) is expressed as follows:

$$\begin{bmatrix} \zeta_d \\ \zeta_q \\ \zeta_0 \end{bmatrix} = \mathbf{T}_p(\theta_e) \begin{bmatrix} \zeta_a \\ \zeta_b \\ \zeta_c \end{bmatrix}, \quad (1)$$

where $\theta_e = p\theta_m$ denotes the electrical rotor angle, with p representing the number of pole pairs, and $\mathbf{T}_p(\theta_e)$ signifies the Park transformation matrix.

In this context, the equations for the stator voltage are articulated as follows.

$$v_d = R_s i_d + \frac{d\lambda_d}{dt} - \omega_e \lambda_q, \quad (2)$$

$$v_q = R_s i_q + \frac{d\lambda_q}{dt} + \omega_e \lambda_d, \quad (3)$$

In this context, R_s denotes the stator resistance, while (i_d, i_q) represent the direct and quadrature axis currents, respectively. The corresponding stator voltages are indicated by (v_d, v_q) , and the flux linkages are denoted as (λ_d, λ_q) . Additionally, the electrical angular velocity is expressed as $\omega_e = p\omega_m$. The flux linkages can be expressed as follows.

$$\lambda_d = L_d i_d + \lambda_f, \quad (4)$$

$$\lambda_q = L_q i_q, \quad (5)$$

where L_d and L_q represent the inductances along the d and q axes, respectively, and λ_f denotes the flux linkage attributed to the permanent magnets. Substituting leads to the classical nonlinear dq model representing the dynamics of the Brushless DC (BLDC) stator Mali and Susitra (2024).

2.2 Torque generation and mechanical subsystem

The expression for the electromagnetic torque is given by:

$$T_e = \frac{3}{2} p [\lambda_f i_q + (L_d - L_q) i_d i_q]. \quad (6)$$

The dynamics of rotor speed adhere to Newton's law of motion:

$$J \frac{d\omega_m}{dt} = T_e - T_L - B\omega_m, \quad (7)$$

where J represents the rotor inertia, B denotes the viscous friction coefficient, ω_m indicates the mechanical angular velocity, and T_L signifies the load torque. Subsection Nonlinear State-Space Formulation In order to enhance the control design process, the dynamics of the Brushless DC (BLDC) are expressed in state-space representation. Let us define the state and input vectors as follows:

$$\mathbf{x}(t) = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} i_d(t) \\ i_q(t) \\ \omega_m(t) \end{bmatrix}, \quad \mathbf{u}(t) = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} v_d(t) \\ v_q(t) \\ T_L(t) \end{bmatrix}, \quad (8)$$

which allows us to express the system as:

$$\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t)) + \mathbf{G}\mathbf{u}(t), \quad (9)$$

where the drift dynamics are given by: $\dot{\mathbf{x}}$

$$\mathbf{f}(\mathbf{x}) = \begin{bmatrix} -\frac{R_s}{L_d} x_1 + \frac{L_q}{L_d} p x_2 x_3 \\ -\frac{R_s}{L_q} x_2 - \frac{L_d}{L_q} p x_1 x_3 - \frac{p\lambda_f}{L_q} x_3 \\ \frac{p}{J} (\lambda_f x_2 + (L_d - L_q) x_1 x_2) - \frac{B}{J} x_3 \end{bmatrix}, \quad (10)$$

and input matrix:

$$\mathbf{G} = \begin{bmatrix} \frac{1}{L_d} & 0 & 0 \\ 0 & \frac{1}{L_q} & 0 \\ 0 & 0 & -\frac{1}{J} \end{bmatrix}. \quad (11)$$

2.3 Control challenges

The nonlinear Brushless DC (BLDC) model presents significant challenges, including the strong interdependence of i_d , i_q , and ω_m , the presence of nonlinear cross-products, and heightened sensitivity to uncertain parameters. Recent studies have highlighted these issues in the context of electromechanical coupling Ye et al. (2023) and the design of robust control systems for high-performance electric vehicle drives Kasri et al. (2024); Cherif et al. (2024). The aforementioned characteristics provide a solid rationale for the implementation of sophisticated methodologies—such as sliding mode control, predictive schemes, and fractional-order strategies—that are fundamentally based on this accurate mathematical model.

3 Design of a filtered high-gain observer for the brushless DC (BLDC) model

The nonlinear Brushless DC (BLDC) model articulated in (Equation 9) forms the basis for the development of a filtered high-gain observer (FHGO). This observer architecture maintains the rapid convergence characteristic of classical high-gain observers (HGOs) while integrating an internal filtering stage that mitigates

their primary limitation: heightened sensitivity to measurement noise. This limitation has been extensively recorded in the literature, where various filtering strategies and adaptive gain mechanisms have been proposed to improve robustness Farza et al. (2021); Mousavi and Guay (2023). Recent developments in FHGO theory have shown notable enhancements compared to traditional HGOs across various applications, such as unmanned aerial vehicles Dam et al. (2024), biological growth processes Robles-Magdaleno et al. (2020), sampled-data systems Hu et al. (2022), electro-hydraulic braking for electric vehicles El-Bakkouri et al. (2023), and motor drive systems Wang et al. (2021). In light of these diverse contexts, a common thread emerges: the implementation of a filter variable alongside an adaptive Riccati-based gain law, which serves to stabilize the estimation process while maintaining swift convergence. Let $\hat{\mathbf{x}} \in \mathbb{R}^3$ represent the estimated state and $\eta \in \mathbb{R}^3$ signify the auxiliary filter variable. Let us define the diagonal scaling matrix as follows: $\Delta = \text{diag}(\varepsilon^{-1}, \varepsilon^{-2}, \varepsilon^{-3})$, where ε is a positive constant, specifically $\varepsilon > 0$. The dynamics of the FHGO are expressed as follows:

$$\dot{\hat{\mathbf{x}}} = f(\hat{\mathbf{x}}) + G\mathbf{u} - \Delta K(t)\eta, \quad \dot{\eta} = -\theta(\eta + C^T(C\hat{\mathbf{x}} - \mathbf{y})), \quad (12)$$

where $\theta > 0$ denotes the cutoff frequency of the internal filter, C represents the output matrix, and $\mathbf{y}$ signifies the measured outputs. The initialization is selected as $\hat{\mathbf{x}}(0) = \mathbf{x}(0)$ and $\eta(0) = \mathbf{0}$ to maintain consistency with the physical system. The observer gain is modified in accordance with the Riccati equation given by

$$\dot{P} = PC^T CP - PF_x P - \beta P, \quad K(t) = P(t)C^T, \quad (13)$$

where $P(0) = P_0 > 0$ is a diagonal and positive definite matrix, $\beta > 0$ represents a forgetting factor, and $F_x = \partial f / \partial \mathbf{x}$ denotes the Jacobian of the system evaluated at $\hat{\mathbf{x}}$. This adaptation mechanism, as inspired by Dam et al. (2024); Robles-Magdaleno et al. (2020), facilitates a dynamic adjustment of the gain in response to local variations within the system, thereby preventing peaking and improving noise immunity.

The incorporation of the filter state η alongside the forgetting factor β establishes a dual mechanism for enhancing robustness: the filter effectively mitigates the transmission of high-frequency components into the high-gain channels, while the Riccati update persistently harmonizes the rate of convergence with the rejection of disturbances El-Bakkouri et al. (2023); Hu et al. (2022). The triangular structure of (12) is notably preserved, yet it undergoes significant simplification when contrasted with the broader matrix-based implementations outlined in Farza et al. (2021). Given that each output possesses a relative degree of one, the anti-shift structure simplifies to the scalar $-\theta$. This reduction decreases computational complexity and enhances implementation in embedded motor drive platforms, where real-time constraints impose stringent limits on the number of arithmetic operations per cycle. Given the standard assumptions regarding bounded system trajectories, Lipschitz continuity of the vector field, and the presence of essentially bounded noise, the FHGO ensures that the estimation error remains uniformly bounded, and exhibits exponential convergence when noise is absent. The stability analysis is performed using a Lyapunov functional that integrates the estimation error, scaled by $P^{-1}(t)$, with the filter state η . The time derivative of this functional is negative semi-definite, which

guarantees robust convergence, as demonstrated in Dam et al. (2024); Mousavi and Guay (2023). While the comprehensive proof is not included in this discussion, the theoretical basis substantiates that the suggested design achieves the intended equilibrium between estimation speed and noise resilience.

It is worth emphasizing that, although observer-based sliding-mode architectures may experience performance degradation when estimation errors are directly injected into discontinuous switching terms, this effect is structurally mitigated in the proposed framework. In particular, the use of a filtered high-gain observer is motivated precisely by its ability to attenuate output-noise propagation, reduce peaking, and improve robustness against high-frequency estimation perturbations, as established in the FHGO literature Ragoubi et al. (2019); Tréangle et al. (2019). Moreover, high-gain observer design in electromechanical feedback systems has shown that an appropriate observer structure is essential to preserve closed-loop performance and avoid excessive sensitivity to measurement noise Khalil (2017). In the proposed observer-controller interconnection, the estimated states are not injected into a pure discontinuous sign function; instead, their influence is further regularized by the filtered sliding surface and the fractional-order reaching dynamics. Therefore, observer-induced mismatch is not assumed to vanish exactly, but rather to remain bounded and to enter the closed-loop dynamics as an attenuated perturbation, which is subsequently handled by the inherent robustness of the fractional-order sliding-mode mechanism. This explains why the proposed FHGO-FOSMC architecture preserves tracking performance without exacerbating chattering.

The FHGO formulation developed for Brushless DC (BLDC) motors incorporates non-peaking characteristics, adaptive Riccati-based gain selection, and internal noise filtering, resulting in a practical and mathematically sound tool for sensorless electric vehicle applications. Furthermore, this observer design is fundamentally aligned with sophisticated control methodologies like fractional-order sliding mode control, wherein dependable state estimation is essential for attaining robustness and optimal performance in electric drive systems.

4 Fractional-order sliding mode control (FOSMC) for the brushless DC (BLDC)

The Brushless DC (BLDC) plant is expressed in a compact form as follows:

$$\dot{\mathbf{x}} = f(\mathbf{x}) + G\mathbf{u}, \quad (14)$$

where $\mathbf{x}^T = [i_d, i_q, \omega_m]$, and the functions $f(\cdot)$ and G are defined in Equations 10, 11. The vector field $f(\cdot)$ is assumed to be locally Lipschitz on a compact set $\Omega \subset \mathbb{R}^3$ containing the closed-loop trajectories generated under bounded references, bounded disturbances, and admissible operating conditions. Only the stator voltages $\mathbf{u}_c^T = [v_d, v_q]$ are controlled; the load torque T_L is introduced through the third input channel of G as a matched disturbance. The previously designed FHGO supplies the currents and speed for feedback, allowing the controller to utilize $\hat{\mathbf{x}}$ alongside a model-based derivative estimate given by

$\hat{\mathbf{x}} = f(\hat{\mathbf{x}}) + G[v_d \ v_q \ 0]^T$, where the disturbance channel is assumed to be zero in the estimate.

Let the reference trajectory be represented as $\mathbf{x}_r(t) = [i_d^*(t), i_q^*(t), \omega^*(t)]^T$, where i_d^* is set to 0 for surface-mounted machines or derived from an MTPA schedule in the salient case, and i_q^* is produced by a speed loop (e.g., proportional–resonant or robust predictive) Cherif et al. (2024); Kasri et al. (2024); ? Let us define the tracking error and its model-based derivative as follows:

$$\mathbf{e} = \mathbf{x}_r - \hat{\mathbf{x}}, \quad \dot{\mathbf{e}} = \dot{\mathbf{x}}_r - \dot{\hat{\mathbf{x}}}, \quad (15)$$

and introduce the composite sliding variable given by

$$\mathbf{s} = \dot{\mathbf{e}} + \Lambda \mathbf{e}, \quad \Lambda = \text{diag}(\lambda_d, \lambda_q, \lambda_\omega), \quad \lambda_i > 0, \quad (16)$$

which enforces second-order error dynamics on the manifold $\mathbf{s} = \mathbf{0}$, thus facilitating simultaneous tracking of both current and speed. By differentiating (Equation 16) and applying (Equation 14) with the estimated signals, we obtain the following expression:

$$\dot{\mathbf{s}} = \ddot{\mathbf{x}}_r - f(\hat{\mathbf{x}}) - G\mathbf{u} + \Lambda \dot{\mathbf{e}} + \Delta(\mathbf{x}, \hat{\mathbf{x}}, T_L), \quad (17)$$

where Δ encompasses the plant/observer mismatch along with the unknown disturbance introduced by T_L . According to the FHGO hypotheses, which include bounded trajectories, local Lipschitz dynamics, and filtered innovations, the term Δ is uniformly bounded. Furthermore, it can be over-bounded by a known constant vector $\bar{\Delta}$, facilitated by rapid estimation and noise attenuation as discussed in the literature Farza et al. (2021); Mousavi and Guay (2023); Dam et al. (2024); El-Bakkouri et al. (2023).

Let $f_c(\hat{\mathbf{x}}) \in \mathbb{R}^2$ represent the electrical components of $f(\hat{\mathbf{x}})$ and define

$$G_c = \text{diag}(L_d^{-1}, L_q^{-1}) \in \mathbb{R}^{2 \times 2}.$$

Partition Equation 17 into its electrical and mechanical channels. In the context of the electrical channel, we have the following equation:

$$\dot{\mathbf{s}}_c = \ddot{\mathbf{x}}_{r,c} - f_c(\hat{\mathbf{x}}) - G_c \mathbf{u}_c + \Lambda_c \dot{\mathbf{e}}_c + \Delta_c, \quad (18)$$

where the subscript $(\cdot)_c$ indicates the $[d, q]$ components. We define $\dot{\mathbf{s}}_c$ using a fractional super-twisting reaching law, which incorporates nonlocal “memory” effects and smooths the discontinuous action, thus minimizing chattering while maintaining robustness Ben Salem et al. (2024a); Gudey et al. (2023); Nicola et al. (2024). For $0 < \beta < 1$, we examine the Caputo-derivative formulation given by

$$D^{\beta+1} \mathbf{s}_c = -K_1 \|\mathbf{s}_c\|^\beta \text{sgn}(\mathbf{s}_c) \quad (19)$$

$-K_2 D^\beta \mathbf{s}_c$, $K_1, K_2 = \text{diag}(k_{1d}, k_{1q}), \text{diag}(k_{2d}, k_{2q}) > 0$, where the integral form (applying I^β componentwise) is expressed as

$$\dot{\mathbf{s}}_c = -K_1 I^\beta (\|\mathbf{s}_c\|^\beta \text{sgn}(\mathbf{s}_c)) - K_2 \mathbf{s}_c. \quad (20)$$

By equating (Equation 18) with (Equation 20), we derive the inner-loop control law

$$\mathbf{u}_c = G_c^{-1} \left[-\ddot{\mathbf{x}}_{r,c} + \Lambda_c \dot{\mathbf{e}}_c + f_c(\hat{\mathbf{x}}) + K_1 I^\beta (\|\mathbf{s}_c\|^\beta \text{sgn}(\mathbf{s}_c)) + K_2 \mathbf{s}_c \right], \quad (21)$$

This formulation captures the fractional reaching dynamics within the current channels while effectively mitigating Δ_c through the mechanism of sliding action. In practice, $I^\beta(\cdot)$ and $D^\beta(\cdot)$ are implemented using a fifth-order Oustaloup approximation across a frequency band that encompasses electrical dynamics and the PWM carrier (e.g., $[10^1, 10^4]$ Hz). This is discretized with bilinear (Tustin) mapping—an approach that has been consistently validated for EV drives integrating DTC/SVM with fractional sliding control Ben Salem et al. (2024a); Le and Hsieh (2024); Nicola et al. (2024).

For implementation purposes, the fractional operator is not regarded as exact, but as a band-limited rational realization obtained through a fifth-order Oustaloup approximation. Accordingly, the mismatch between the ideal fractional operator and its finite-order realization can be interpreted as an additional bounded approximation error, which is absorbed into the residual perturbation channel of the closed-loop analysis. The selected frequency range was chosen so as to cover the dominant mechanical/sliding dynamics at the low-frequency end and the relevant electrical/control bandwidth of the Brushless DC (BLDC) drive at the high-frequency end, without unnecessarily extending the approximation toward ultra-high-frequency components that are not meaningful under practical actuation and sampling constraints. From the implementation viewpoint, the fifth-order realization represents a compromise between approximation fidelity and digital tractability: increasing the approximation order or enlarging the design band may improve matching accuracy, but it also increases filter order, discretization sensitivity, and computational burden. Therefore, the adopted fifth-order Oustaloup realization was retained as a balanced choice for practical embedded implementation Baranowski et al. (2015); Tepljakov et al. (2020); Baranowski et al. (2016). The mechanical channel is managed in a cascade configuration through the torque map: a speed loop produces i_q^* with a bandwidth significantly lower than that of the electrical loop (on the order of tens of hertz). Proportional–resonant or robust predictive implementations align well with HIL-validated Brushless DC (BLDC) traction results Kasri et al. (2024); Cherif et al. (2024). The d -axis reference i_d^* is established at zero for surface-mounted machines or derived from an MTPA/MTPV profile in the presence of saliency. *Tuning guidelines.* Select $\Lambda_c = \text{diag}(\lambda_d, \lambda_q)$ to position the electrical closed-loop poles within the frequency range of 200–400 Hz; choose λ_ω in the range of 20–40 Hz to ensure bandwidth separation. Select $\beta \in [0.6, 0.9]$ to achieve a balance between memory and convergence, where a larger β results in smoother actions and reduced high-frequency content Ben Salem et al. (2024a); Gudey et al. (2023). The gains K_1 and K_2 should exceed the uncertainty bound $\bar{\Delta}_c$: initiate by configuring K_2 to achieve the intended linear decay of $\mathbf{s}_c$, subsequently adjust K_1 to reduce the sliding neighborhood while avoiding any perceptible switching. Implement saturation and rate limits on v_d and v_q in accordance with the DC-link, SVM modulation, and inverter constraints (Le and Hsieh (2024); ?)Improved discussion on composite DTC couplings.

Remark on Stability. Utilizing the FHGO properties, which include uniformly bounded estimation error and non-peaking behavior, alongside standard fractional Lyapunov arguments for (Equation 4), it can be established that $\mathbf{s}_c$ approaches a small neighborhood of the

origin within a finite (fractional) time frame and remains bounded. Furthermore, on the manifold, the equation $\dot{\mathbf{e}}_c + \Lambda_c \mathbf{e}_c = \mathbf{0}$ guarantees exponential tracking of the current. The speed loop demonstrates robustness through the separation of bandwidth. The conclusions drawn align with the findings from experimental and HIL reports concerning Brushless DC (BLDC)/IM/SRM drives that utilize fractional/SMC/DTC hybrids Ben Salem et al. (2024a); Gudey et al. (2023); Le and Hsieh (2024); Nicola et al. (2024).

4.1 Deterioration of sliding mode configuration in the integer-order regime

An enhanced structural understanding of the proposed formulation emerges upon investigating the integer-order limit $\beta = 1$. The general control law is expressed as follows:

$$v_q = L_q \left(-\ddot{\omega}_m^* + f_q(x) + k_1 I^\beta (|s_f|^{1/\beta} \text{sgn}(s_f)) + k_2 s_f \right), \quad (22)$$

where the filtered sliding surface is defined as

$$s_f = \dot{e} + \lambda e, \quad e = \omega_m^* - \omega_m. \quad (23)$$

Integer-order limit ($\beta = 1$).

When $\beta = 1$, the nonlinear power term simplifies to

$$|s_f|^{1/\beta} \text{sgn}(s_f) = |s_f| \text{sgn}(s_f) = s_f, \quad (24)$$

thus, the fractional operator is equivalent to the classical integral. Consequently, the control law is expressed as

$$v_q = L_q \left(-\ddot{\omega}_m^* + f_q(x) + k_1 \int_0^t s_f(\tau) d\tau + k_2 s_f \right). \quad (25)$$

In this configuration, the induced sliding dynamics is described by the following equation:

$$\ddot{s}_f + k_2 \dot{s}_f + k_1 s_f = 0, \quad (26)$$

which represents a linear second-order differential equation. The nonlinear reaching term is eliminated, resulting in a smooth and globally Lipschitz vector field.

Consequently, the fundamental geometric property of sliding mode control—namely, the creation of an invariant manifold enforced by a nonlinear dissipation inequality of the form

$$s_f \dot{s}_f < 0 \quad \text{through a power – type or discontinuous term,} \quad (27)$$

is lost. Structural Interpretation.

It is crucial to note that the sliding surface itself,

$$s_f = \dot{e} + \lambda e, \quad (28)$$

already signifies a combination of proportional and derivative elements within the error coordinates. When $\beta = 1$, the additional integral term $\int s_f$ contributes to an enhanced accumulation effect. Consequently, the closed-loop configuration is analogous to a linear regulator operating on transformed coordinates, structurally similar to an extended PID controller.

Consequently, the integer-order limit fails to align with traditional sliding mode control methodologies. Instead, it signifies a decline of the nonlinear sliding mechanism into a linear integral regulation law characterized by the surface variable. Fractional regime ($0 < \beta < 1$).

For $0 < \beta < 1$, the exponent $1/\beta > 1$ reintroduces a superlinear power term,

$$\phi(s_f) = |s_f|^{1/\beta} \text{sgn}(s_f), \quad (29)$$

which reshapes the vector field near the manifold $s_f = 0$. The dissipation exhibits a state-dependent and nonlinear nature, leading to a generalized sliding behavior that is marked by non-quadratic energy decay and increased robustness.

Consequently, the parameter β plays a crucial role in determining a structural transition: at $\beta = 1$, the sliding mode characteristic ceases to exist, and the controller functions as a linear PI-type regulator in surface coordinates; for the range $0 < \beta < 1$, the nonlinear sliding geometry is reinstated via fractional superlinear dissipation.

A notable aspect of the proposed formulation is its structural adaptability via the fractional parameter β . As β approaches 1, the controller functions as a smooth PI-type regulator that operates on the sliding surface, delivering gentle corrections and requiring minimal actuation effort during nominal operating conditions with minor disturbances. As β diminishes, the nonlinear power term $|s|^{1/\beta} \text{sgn}(s)$ gradually alters the dissipation mechanism, enhancing robustness and the ability to reject disturbances. Specifically, when $\beta = 0.5$, the resulting configuration exhibits characteristics akin to a fractional super-twisting scheme. Conversely, as β approaches 0, there is potential for achieving higher-order sliding mode behavior. This continuous deformation allows for a singular control architecture to shift from smooth integral regulation to highly nonlinear robust sliding dynamics, providing a cohesive framework that can adjust its aggressiveness and robustness in response to the level of perturbation.

5 Joint stability of FHGO and FOSMC

In this section, the closed-loop interconnection between the filtered high-gain observer (FHGO) and the fractional-order sliding-mode controller (FOSMC) is analyzed more explicitly. The purpose is twofold: first, to make the Lyapunov-based derivation more transparent; and second, to clarify how the practical approximation error of the fractional operator can be incorporated into the stability argument.

Let $\mathbf{x} \in \mathbb{R}^3$ denote the actual Brushless DC (BLDC) state, $\hat{\mathbf{x}} \in \mathbb{R}^3$ the FHGO estimate, and define the estimation error

$$\tilde{\mathbf{x}} = \mathbf{x} - \hat{\mathbf{x}}.$$

Let $\mathbf{s} \in \mathbb{R}^3$ denote the sliding variable introduced in (Equation 16), and stack the relevant closed-loop variables as

$$\zeta = \begin{bmatrix} \tilde{\mathbf{x}} \\ \mathbf{s} \end{bmatrix} \in \mathbb{R}^6, \quad \zeta^\top = [\tilde{\mathbf{x}}^\top \quad \mathbf{s}^\top].$$

The analysis is carried out under the following assumptions.

- $f(\cdot)$ is locally Lipschitz on the operating domain of interest, i.e., there exists $L_f > 0$ such that

$$\|f(\mathbf{x}) - f(\hat{\mathbf{x}})\| \leq L_f \|\tilde{\mathbf{x}}\|.$$

- The measurement equation is $\mathbf{y} = \mathbf{x} + \mathbf{n}$, with $C = I_3$, and the sensor noise is essentially bounded:

$$\|\mathbf{n}(t)\| \leq n_{\max}.$$

- The matched disturbance associated with load torque and unmodeled input-channel effects is bounded:

$$\|\mathbf{d}(t)\| \leq d_{\max}.$$

- The practical realization of the fractional operator is obtained through a finite-order Oustaloup approximation over a prescribed frequency band. Let $\varepsilon_\beta(t)$ denote the corresponding approximation mismatch between the ideal fractional operator and its finite-order realization. It is assumed that

$$\|\varepsilon_\beta(t)\| \leq \varepsilon_{\beta,\max}.$$

Under these assumptions, the approximation mismatch can be incorporated into the perturbation channel of the sliding dynamics.

5.1 Estimation-error and filter dynamics

From the plant dynamics and the FHGO Equation 12, the estimation error satisfies

$$\begin{aligned} \dot{\tilde{\mathbf{x}}} &= \dot{\mathbf{x}} - \dot{\hat{\mathbf{x}}} \\ &= \mathbf{f}(\mathbf{x}) + \mathbf{G}\mathbf{u} + \mathbf{d}(t) - (\mathbf{f}(\hat{\mathbf{x}}) + \mathbf{G}\mathbf{u} - \Delta\mathbf{K}\eta) \\ &= \Delta\mathbf{f} + \Delta\mathbf{K}\eta + \mathbf{d}(t), \end{aligned} \quad (30)$$

where

$$\Delta\mathbf{f} := \mathbf{f}(\mathbf{x}) - \mathbf{f}(\hat{\mathbf{x}}), \quad \|\Delta\mathbf{f}\| \leq L_f \|\tilde{\mathbf{x}}\|.$$

Using $C = I_3$ and $\mathbf{y} = \mathbf{x} + \mathbf{n}$, the internal filter state obeys

$$\begin{aligned} \dot{\eta} &= -\theta(\eta + (\hat{\mathbf{x}} - \mathbf{x}) - \mathbf{n}) \\ &= -\theta\eta + \theta\tilde{\mathbf{x}} + \theta\mathbf{n}. \end{aligned} \quad (31)$$

5.2 Sliding-channel dynamics with fractional approximation error

In the ideal formulation, the sliding dynamics under the proposed law are written as

$$\dot{\mathbf{s}} = -K_1 I^\beta \left(\|\mathbf{s}_c\|^{\frac{1}{\beta}} \text{sgn}(\mathbf{s}_c) \right) - K_2 \mathbf{s}_c + \begin{bmatrix} \Delta_c \\ \Delta_\omega \end{bmatrix}.$$

However, in the practical implementation, $I^\beta(\cdot)$ is replaced by its finite-order approximation. Therefore, the actual sliding dynamics can be written more precisely as

$$\dot{\mathbf{s}} = -K_1 I^\beta \left(\|\mathbf{s}_c\|^{\frac{1}{\beta}} \text{sgn}(\mathbf{s}_c) \right) - K_2 \mathbf{s}_c + \begin{bmatrix} \Delta_c^\star \\ \Delta_\omega \end{bmatrix}, \quad (32)$$

where

$$\Delta_c^\star := \Delta_c + \varepsilon_\beta(t),$$

and consequently

$$\|\Delta_c^\star\| \leq \bar{\Delta}_c + \varepsilon_{\beta,\max}.$$

Hence, the finite-order approximation does not alter the structure of the proof; it only enlarges the bounded perturbation channel affecting the sliding subsystem.

5.3 Lyapunov candidate

To capture the coupled effect of state-estimation error, sliding dynamics, and observer filtering, consider the Lyapunov function

$$V(\zeta, \eta) = \tilde{\mathbf{x}}^\top P^{-1} \tilde{\mathbf{x}} + \frac{1}{2} \mathbf{s}^\top K^{-1} \mathbf{s} + \frac{\rho}{2} \eta^\top \eta, \quad \rho > 0, \quad (33)$$

where $P(t) > 0$ is the Riccati state of the FHGO and $K = \text{diag}(k_d, k_q, k_\omega) > 0$. Since P^{-1} , K^{-1} , and ρI are positive definite, there exist $\underline{\lambda}, \bar{\lambda} > 0$ such that

$$\underline{\lambda} (\|\tilde{\mathbf{x}}\|^2 + \|\mathbf{s}\|^2 + \|\eta\|^2) \leq V \leq \bar{\lambda} (\|\tilde{\mathbf{x}}\|^2 + \|\mathbf{s}\|^2 + \|\eta\|^2). \quad (34)$$

5.4 Time derivative of the Lyapunov function

The derivative of the estimation term is

$$\frac{d}{dt} (\tilde{\mathbf{x}}^\top P^{-1} \tilde{\mathbf{x}}) = 2\tilde{\mathbf{x}}^\top P^{-1} \dot{\tilde{\mathbf{x}}} + \tilde{\mathbf{x}}^\top \dot{P}^{-1} \tilde{\mathbf{x}}.$$

Using $\dot{P}^{-1} = -P^{-1} \dot{P} P^{-1}$ together with (Equation 13), one obtains

$$\dot{P}^{-1} = -C^\top C P + ? \quad (35)$$

More precisely, since

$$\dot{P} = P C^\top C P - P F_x P - \beta P,$$

it follows that

$$\begin{aligned} \dot{P}^{-1} &= -P^{-1} (P C^\top C P - P F_x P - \beta P) P^{-1} \\ &= -C^\top C + F_x + \beta P^{-1}. \end{aligned} \quad (36)$$

With $C = I_3$ (Equation 30), and (Equation 36),

$$\begin{aligned} \frac{d}{dt} (\tilde{\mathbf{x}}^\top P^{-1} \tilde{\mathbf{x}}) &= 2\tilde{\mathbf{x}}^\top P^{-1} (\Delta\mathbf{f} + \Delta\mathbf{K}\eta + \mathbf{d}) + \tilde{\mathbf{x}}^\top (-I_3 + F_x + \beta P^{-1}) \tilde{\mathbf{x}} \\ &\leq -\alpha_x \|\tilde{\mathbf{x}}\|^2 + 2\|P^{-1} \Delta\mathbf{K}\| \|\tilde{\mathbf{x}}\| \|\eta\| + 2\lambda_{\max}(P^{-1}) \|\tilde{\mathbf{x}}\| \|\mathbf{d}\|, \end{aligned} \quad (37)$$

where $\alpha_x > 0$ is obtained provided the observer design parameters are chosen such that the negative quadratic contribution dominates the Lipschitz and Riccati residual terms.

The derivative of the filter-energy term is

$$\begin{aligned} \frac{d}{dt} \left(\frac{\rho}{2} \eta^\top \eta \right) &= \rho \eta^\top \dot{\eta} \\ &= -\rho \theta \|\eta\|^2 + \rho \theta \eta^\top \tilde{\mathbf{x}} + \rho \theta \eta^\top \mathbf{n} \\ &\leq -\frac{\rho \theta}{2} \|\eta\|^2 + \frac{\rho \theta}{2} \|\tilde{\mathbf{x}}\|^2 + \frac{\rho \theta}{2} n_{\max}^2. \end{aligned} \quad (38)$$

For the sliding term, using (Equation 32),

$$\begin{aligned} \frac{d}{dt} \left(\frac{1}{2} \mathbf{s}^\top K^{-1} \mathbf{s} \right) &= \mathbf{s}^\top K^{-1} \dot{\mathbf{s}} \\ &= -\mathbf{s}_c^\top K_1^\# I^\beta \left(\|\mathbf{s}_c\|^{\frac{1}{\beta}} \text{sgn}(\mathbf{s}_c) \right) - \mathbf{s}_c^\top K_2^\# \mathbf{s}_c \\ &\quad + \lambda_{\max}(K^{-1}) \|\mathbf{s}\| (\|\Delta_c^\star\| + \|\Delta_\omega\|), \end{aligned} \quad (39)$$

where $K_i^\# := K^{-1} K_i > 0$.

By the standard fractional dissipation inequality used in high-order fractional sliding-mode analysis, there exists $c_\beta > 0$ such that

$$\mathbf{s}_c^\top I^\beta \left(\|\mathbf{s}_c\|^{\frac{1}{\beta}} \text{sgn}(\mathbf{s}_c) \right) \geq c_\beta \|\mathbf{s}_c\|^{1+\frac{1}{\beta}}. \quad (40)$$

Substituting (Equation 40) into (Equation 39) yields

$$\frac{d}{dt} \left(\frac{1}{2} \mathbf{s}^\top K^{-1} \mathbf{s} \right) \leq -c_\beta \lambda_{\min}(K_1^\#) \|\mathbf{s}_c\|^{1+\frac{1}{\beta}} - \lambda_{\min}(K_2^\#) \|\mathbf{s}_c\|^2 + \lambda_{\max}(K^{-1}) \|\mathbf{s}\| (\bar{\Delta}_c + \varepsilon_{\beta, \max} + \|\Delta_\omega\|). \quad (41)$$

5.5 Absorption of mixed terms and final differential inequality

The mixed products in (Equation 37) are handled by Young's inequality. Let

$$a_\Delta := \|P^{-1} \Delta K\|.$$

Then

$$2a_\Delta \|\tilde{\mathbf{x}}\| \|\eta\| \leq \varepsilon_\eta \|\tilde{\mathbf{x}}\|^2 + \frac{a_\Delta^2}{\varepsilon_\eta} \|\eta\|^2, \quad (42)$$

$$2\lambda_{\max}(P^{-1}) \|\tilde{\mathbf{x}}\| \|d\| \leq \varepsilon_d \|\tilde{\mathbf{x}}\|^2 + \frac{\lambda_{\max}^2(P^{-1})}{\varepsilon_d} d_{\max}^2. \quad (43)$$

Likewise, the perturbation term in (Equation 41) is bounded componentwise by quadratic terms in $\mathbf{s}$ plus constants depending on $\bar{\Delta}_c$, $\varepsilon_{\beta, \max}$, and $\bar{\Delta}_\omega$. Therefore, collecting (Equations 37, 38, 41–43), there exist strictly positive constants $c_x, c_\eta, c_{1+1/\beta}, c_2, c_\omega, \Gamma_1, \Gamma_2, \Gamma_3, \Gamma_4, \Gamma_5$ such that

$$\begin{aligned} \dot{V} \leq & -c_x \|\tilde{\mathbf{x}}\|^2 - c_\eta \|\eta\|^2 - c_{1+1/\beta} \|\mathbf{s}_c\|^{1+\frac{1}{\beta}} - c_2 \|\mathbf{s}_c\|^2 - c_\omega s_\omega^2 + \Gamma_1 d_{\max}^2 \\ & + \Gamma_2 n_{\max}^2 + \Gamma_3 \bar{\Delta}_c^2 + \Gamma_4 \bar{\Delta}_\omega^2 + \Gamma_5 \varepsilon_{\beta, \max}^2. \end{aligned} \quad (44)$$

Since V is quadratically equivalent to $\|\tilde{\mathbf{x}}\|^2 + \|\eta\|^2 + \|\mathbf{s}\|^2$ by (Equation 34), the differential inequality (Equation 44) implies the existence of $\alpha > 0$ and $\gamma > 0$ such that

$$\dot{V} \leq -\alpha V + \gamma (d_{\max}^2 + n_{\max}^2 + \varepsilon_{\beta, \max}^2). \quad (45)$$

5.6 Stability interpretation

Applying Grönwall's lemma to (Equation 45), one obtains

$$V(t) \leq V(0)e^{-\alpha t} + \frac{\gamma}{\alpha} (d_{\max}^2 + n_{\max}^2 + \varepsilon_{\beta, \max}^2) (1 - e^{-\alpha t}). \quad (46)$$

Therefore, two cases follow naturally.

- Ideal nominal case: if $d_{\max} = 0$, $n_{\max} = 0$, and $\varepsilon_{\beta, \max} = 0$, then

$$V(t) \leq V(0)e^{-\alpha t},$$

which implies exponential convergence

$$\tilde{\mathbf{x}}(t) \rightarrow 0, \quad \eta(t) \rightarrow 0, \quad \mathbf{s}(t) \rightarrow 0.$$

- Practical implementation case: if the load disturbance, measurement noise, and fractional approximation mismatch are bounded but nonzero, then the trajectories remain ultimately bounded, with residual radius proportional to

$$\sqrt{d_{\max}^2 + n_{\max}^2 + \varepsilon_{\beta, \max}^2}.$$

Hence the closed-loop system is naturally interpreted in the ISS/practical stability sense.

This refined interpretation addresses both the observer–controller coupling and the practical finite-order realization of the fractional operator. Accordingly, the ideal theoretical design preserves exponential convergence in the nominal case, whereas the implementable closed-loop architecture retains bounded robust convergence under realistic perturbations and approximation-induced residuals.

While the current investigation lacks experimental validation, the theoretical assurances offered by the Lyapunov-based stability analysis, in conjunction with the integration of realistic plant perturbations (such as thermal drift, Coulomb friction, rolling resistance, and aerodynamic load), affirm that the numerical framework effectively encapsulates the primary nonlinear dynamics of practical Brushless DC (BLDC) traction systems. The interconnection between the observer and controller is thoroughly examined in the presence of bounded disturbances, with all simulations performed on a nonlinear dq model that is well-established in the field of electric drive research. Consequently, the conclusions reached are fundamentally based on system dynamics instead of artifacts specific to implementation, thereby creating a robust theoretical and numerical basis for subsequent hardware validation.

6 Methodology and simulation framework

This section provides a comprehensive overview of the numerical methodology employed to validate the proposed filtered high-gain observer (FHGO) and fractional-order sliding mode control (FOSMC) strategy. The simulation environment, design of the reference signal, implementation of the controller, selection of parameters, and criteria for performance evaluation are thoroughly documented to guarantee reproducibility and clarity.

6.1 Simulation environment and numerical setup

All numerical experiments were conducted utilizing MATLAB/Simulink 2026 in the context of a continuous-time simulation framework. This environment was chosen because of its extensive use in both academic and industrial research concerning electric drive systems, along with its proven capability in managing nonlinear dynamics, observer-based estimation methods, and sophisticated control strategies. The entire closed-loop architecture, which encompasses the nonlinear Brushless DC (BLDC) motor model articulated in the dq reference frame, the filtered high-gain observer (FHGO), the fractional-order sliding mode controller (FOSMC), reference trajectory generators, and performance evaluation blocks, was realized in Simulink through the integration of standard library components alongside custom MATLAB Function blocks. The simulation setup employed mirrors the operating conditions characteristic of a *typical electric vehicle traction motor*, specifically a medium-power surface-mounted Brushless DC (BLDC) machine frequently utilized in passenger electric cars and light electric vehicles. Specifically, the motor parameters, reference speed profiles, and controller bandwidths were chosen to replicate realistic scenarios of

acceleration, cruising, and deceleration that are typically observed in urban driving cycles. This design decision guarantees that the results achieved are not merely theoretical; rather, they offer valuable insights into the anticipated performance of the proposed control strategy in conditions that are practically significant. A variable-step numerical integration method was utilized to effectively capture the nonlinear characteristics, rapid transient phenomena, and non-smooth behaviors associated with sliding mode control and fractional-order operators. Specifically, the Dormand–Prince method (ode45), which is an explicit Runge–Kutta solver of order 45, was chosen for its robustness and efficiency in addressing stiff-like dynamics that emerge from high-gain observers and discontinuity-regularized control laws.

The configuration of the solver was established in the following manner.

- Simulation horizon: $t \in [0, 5.4]$ s, which includes the phases of acceleration, steady-state, and deceleration.
- Category of Solver: Variable-step explicit Runge–Kutta method (Dormand–Prince).
- Relative tolerance: 10^{-3} , which maintains an equilibrium between numerical accuracy and computational expense.
- Absolute tolerance: This parameter is adjusted automatically by the solver.
- Detection of Zero-Crossings: Activated in a non-adaptive configuration to effectively handle switching-like dynamics while avoiding artificial chattering.

This numerical configuration was chosen to facilitate equitable and consistent comparisons between classical PID control, integer-order sliding mode control, and fractional-order sliding mode control, all while preserving numerical stability and preventing artifacts introduced by the solver. The variations in observed performance can thus be primarily ascribed to the control strategy, rather than to the effects of discretization. This configuration offers a well-considered compromise between numerical precision and computational efficiency, facilitating thorough comparative analysis. As illustrated in Figure ?? the numerical methodology employed in this study is succinctly summarized. The sliding surface undergoes initial filtering via a stable first-order dynamic to enhance numerical robustness before the implementation of both integer- and fractional-order operators. The evaluation framework consists of two distinct experimental stages. In Stage 1, a traditional PID controller is evaluated alongside a fractional sliding mode controller with a fixed order of $\beta = 0.6$. This comparison utilizes standard tracking indices, specifically RMSE, IAE, and ITAE, in addition to a metric for control smoothness. In Stage 2, a sensitivity analysis concerning the fractional order β is conducted. At this stage, the classical tracking indices (RMSE, IAE, and ITAE) are maintained to facilitate a fair comparison among controllers. Additionally, new deviation-based and sliding-surface indices are incorporated to assess the influence of fractional dynamics on robustness and the smoothness of control.

6.2 Brushless DC (BLDC) motor parameters

The parameters of the Brushless DC (BLDC) motor utilized in the simulations pertain to a medium-power electric vehicle traction motor and are outlined as follows:

$$\begin{aligned} R_s &= 0.30 \, \Omega, \\ L_d &= L_q = 8 \times 10^{-4} \, \text{H}, \\ \lambda_f &= 0.06 \, \text{Wb}, \\ p &= 4, \\ J &= 2.5 \times 10^{-3} \, \text{kg m}^2, \\ B &= 1.2 \times 10^{-4} \, \text{N m s/rad}. \end{aligned} \quad (47)$$

The parameters were maintained at a constant level throughout all experiments to effectively isolate the impact of the control strategy and fractional order on the performance of the system.

To assess robustness within realistic operating conditions, the Brushless DC (BLDC) plant integrates structured nonlinear perturbations. Initially, a temperature-dependent parametric uncertainty is incorporated by permitting the stator resistance and permanent magnet flux linkage to fluctuate with the squared quadrature current, specifically $R_s = R_{s0}(1 + k_{R_s}i_q^2)$ and $\lambda_f = \lambda_{f0}(1 - k_{\lambda}i_q^2)$, which simulates thermal effects and flux weakening. Secondly, the load torque comprises rolling and aerodynamic components expressed as $T_{L,\text{speed}} = T_{\text{roll}} + k_{\text{aero}}\omega_m^2$, which denote speed-dependent mechanical losses characteristic of electric vehicle traction. Third, smooth Coulomb friction is represented by the expression $T_c \tanh(\omega_m/\omega_0)$, which incorporates nonlinear dissipation in the vicinity of zero speed. The perturbations lead to a load structure that is nonlinear, state-dependent, and speed-dependent, thereby ensuring that the performance of the controller is validated under conditions that extend beyond nominal operating parameters.

It should be emphasized that the numerical validation is not limited to a single nominal simulation condition. The proposed controller is assessed under a time-varying acceleration–cruise–deceleration profile, observer-based feedback with measurement noise and initialization mismatch, temperature-dependent parameter variations, nonlinear Coulomb friction, and speed-dependent rolling and aerodynamic load effects. In addition, the Stage II analysis examines multiple fractional orders over the same nonlinear perturbed plant, thereby providing a second validation layer for robustness, smoothness, and structural energy behavior. This type of structured non-ideal validation is also consistent with recent fractional-order motor-control studies reported in the literature [Bian and Chien \(2025\)](#); [Ge and Liu \(2022\)](#).

6.3 Design of reference trajectory

The mechanical speed reference $\omega_m^*(t)$ was produced utilizing a fifth-order polynomial trajectory, thereby guaranteeing the continuity of the reference signal along with its first two derivatives. This decision holds significant importance for controllers based on sliding mode, as the control law explicitly incorporates higher-order derivatives.

The reference trajectory is expressed as follows:

$$\omega_m^*(t) = \begin{cases} a_0 + a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 + a_5 t^5, & t \in [0, T_a], \\ \omega_{\text{ref}}, & t \in (T_a, T_d), \\ b_0 + b_1 t + b_2 t^2 + b_3 t^3 + b_4 t^4 + b_5 t^5, & t \in [T_d, T], \end{cases} \quad (48)$$

where the coefficients a_i and b_i are determined to meet the boundary conditions for position, velocity, and acceleration. This profile simulates the phases of acceleration, cruising, and deceleration that are characteristic of electric vehicle operation.

6.4 Parameter selection of the FHGO

The filtered high-gain observer (FHGO) underwent numerical validation under complex initial conditions and the presence of measurement noise to evaluate its convergence, robustness, and ability to suppress peaking effects.

The plant states were initialized at $\mathbf{x}(0) = \mathbf{0}$, whereas the observer was deliberately initialized with an estimation mismatch, given by

$$\hat{\mathbf{x}}(0) = [0.5 \quad -0.3 \quad 20]^T, \quad \boldsymbol{\eta}(0) = \mathbf{0}, \quad P(0) = P_0 \succ 0.$$

This ensures the excitation of the dynamics associated with the estimation error.

Measurement noise was introduced as

$$\mathbf{y}(t) = \mathbf{x}(t) + \mathbf{n}(t),$$

where $\mathbf{n}(t)$ is modeled as bounded Gaussian noise.

The observer scaling parameter was chosen to be $\varepsilon = 0.1$, effectively balancing the need for rapid convergence with the necessity to prevent excessive peaking. The internal filter cutoff frequency was established at $\theta = 10$, which meets the condition $\theta > 1/\varepsilon$ and guarantees adequate suppression of high-frequency components. The Riccati forgetting factor was selected as $\beta = 5$ to ensure stable adaptive gain evolution while avoiding excessive bandwidth variation.

The parameters of the proposed FHGO–FOSMC scheme were not selected arbitrarily, but according to a structured tuning rationale. The FHGO parameters were chosen from the classical trade-off between rapid estimation and attenuation of peaking/noise effects: ε sets the observer scaling, θ determines the internal filtering bandwidth, and the Riccati forgetting factor regulates gain adaptation. The filtered-surface parameter α was selected as a compromise between numerical smoothing and transient responsiveness. The sliding-surface gains in Λ were chosen to impose the desired error-convergence rate while preserving bandwidth separation between electrical and mechanical dynamics. The control gains K_1 and K_2 were selected to dominate the bounded uncertainty term without inducing unnecessarily aggressive actuation. Finally, the fractional exponent β was assessed systematically through the Stage II sensitivity analysis, where its influence on tracking performance, control smoothness, control energy, and structural deformation was explicitly quantified.

6.5 Definition of tracking error and sliding surface

Define the mechanical speed tracking error as follows:

$$e_q(t) = \omega_m^*(t) - \omega_m(t), \quad (49)$$

where $\omega_m^*(t)$ represents the reference speed trajectory and $\omega_m(t)$ indicates the actual rotor speed. The time derivative $\dot{e}_q(t)$ is presumed to be accessible via the observer-based estimation framework delineated in the preceding sections.

The sliding variable is formulated in a conventional geometric manner as

$$s_q(t) = \dot{e}_q(t) + \lambda_q e_q(t), \quad (50)$$

where $\lambda_q > 0$ serves as a design parameter that dictates the intended convergence rate of the tracking error dynamics. The manifold $s_q(t) = 0$ establishes a first-order stable error dynamics described by the equation $\dot{e}_q + \lambda_q e_q = 0$, which results in exponential convergence within the ideal sliding regime.

It is crucial to highlight that (Equation 50) solely delineates the intended sliding manifold and does not yet articulate the control law. The control input is formulated in the following sections by manipulating the progression of $s_q(t)$ via integer- and fractional-order sliding mode reaching laws.

In practical numerical implementations, the direct application of fractional differentiation or integration to the sliding variable $s_q(t)$ can result in the amplification of high-frequency components, numerical stiffness, and diminished robustness of the simulation, especially when observer dynamics and nonlinear switching-like terms are present. In order to address these challenges, the sliding variable undergoes processing via a first-order stable filter before being incorporated into the control law.

In particular, the filtered sliding variable $s_f(t)$ is characterized by the following equation:

$$\dot{s}_f(t) = -\alpha s_f(t) + \alpha s_q(t), \quad (51)$$

where $\alpha > 0$ serves as a design constant. The transfer function in question is given by

$$S_f(s) = \frac{\alpha}{s + \alpha} S_q(s), \quad (52)$$

demonstrating that it is strictly proper and asymptotically stable. In this investigation, $\alpha = 1$ was chosen as a balance between numerical smoothing and rapid transient response.

The filtering stage maintains the integrity of the sliding manifold definition, as $s_f(t) = s_q(t)$ holds true at steady state. However, it introduces an auxiliary dynamics that is exponentially stable, thereby regularizing the signal before it undergoes fractional processing.

The incorporation of the first-order filter (Equation 51) does not alter the equilibrium point or the sliding manifold of the closed-loop system. Rather, it enhances the original dynamics by incorporating an additional linear and exponentially stable subsystem that is influenced by the sliding variable $s_q(t)$.

From (Equation 51), the filtering dynamics is input-to-state stable concerning $s_q(t)$, and satisfies

$$\|s_f(t)\| \leq e^{-\alpha t} \|s_f(0)\| + \sup_{0 \leq \tau \leq t} \|s_q(\tau)\|. \quad (53)$$

Consequently, there exists a constant $c > 0$ such that

$$\|s_f(t)\| \leq c \|s_q(t)\|, \quad (54)$$

for all $t \geq 0$.

Consequently, any Lyapunov function that is initially expressed in relation to the sliding variable $s_q(t)$ can be suitably extended to the filtered variable $s_f(t)$ without any loss of generality. The filtering stage incorporates an auxiliary dynamics that is exponentially stable, thereby maintaining the equilibrium and robustness characteristics of the closed-loop system. As a result, the findings regarding exponential stability and input-to-state stability (ISS) for the unfiltered scenario continue to hold true.

In the initial examination, the stability of the closed-loop system was defined using the composite Lyapunov function

$$V = V_{\text{obs}} + V_s, \quad (55)$$

where V_{obs} denotes the energy of the observer estimation error and V_s encapsulates the sliding-mode tracking dynamics related to s_q .

To explicitly incorporate the filtering dynamics defined in (Equation 51), we consider the extended Lyapunov function given by

$$V_{\text{ext}} = V + \frac{\alpha}{2} \|s_f - s_q\|^2, \quad (56)$$

where $\alpha > 0$ and $s_f - s_q$ represents the filtering error.

By differentiating (Equation 56) with respect to time, we obtain

$$\dot{V}_{\text{ext}} = \dot{V} + \alpha(s_f - s_q)(\dot{s}_f - \dot{s}_q). \quad (57)$$

From the filter dynamics, we have the following equation:

$$\dot{s}_f - \dot{s}_q = -\alpha(s_f - s_q), \quad (58)$$

indicating that the system is exponentially stable. Substituting into (Equation 57) yields

$$\dot{V}_{\text{ext}} = \dot{V} - \alpha^2 \|s_f - s_q\|^2. \quad (59)$$

Consequently, the extended Lyapunov derivative maintains the integrity of the original dissipation inequality while incorporating supplementary negative-definite damping. The filtering stage functions exclusively as a numerically stabilizing regularization mechanism, without altering the asymptotic convergence guarantees of the proposed control scheme.

6.6 Performance indices

In order to conduct a thorough assessment of the effects of fractionalization, the controller characterized by $\beta = 1$ (integer-order sliding formulation) is selected as the foundational reference case. All fractional configurations ($0 < \beta < 1$) are quantitatively assessed in relation to this baseline.

Define the tracking error as follows:

$$e_q(t) = \omega_{\text{ref}}(t) - \omega_m(t), \quad (60)$$

where ω_{ref} denotes the reference angular velocity and ω_m represents the measured rotor speed.

Furthermore, to measure the deviation caused by fractionalization in the control action, we introduce the control deviation error:

$$e_u(t) = u_{\beta=1}(t) - u_\beta(t), \quad (61)$$

This quantifies the extent to which the fractional-order control law deviates from the integer-order scenario.

All indices are calculated over a finite interval $[0, T]$.

This assessment framework facilitates a comprehensive and multidimensional comparison. In addition to RMSE, IAE, and ITAE, which measure tracking accuracy, the indices J_u and $J_{\dot{u}}$ evaluate the demand for actuation and the smoothness of the response. The deviation-based indices (J_{e_u} , ITAE_{e_u} , and J_s) serve to explicitly quantify the structural transformation brought about by fractionalization in relation to the reference case where $\beta = 1$. As a result, enhancements in performance are assessed not only through the lens of tracking precision but also with regard to the reconfiguration of control, the fluidity of operation, and alterations in dynamical behavior.

The adopted performance indicators serve two complementary purposes. On the one hand, RMSE, IAE, and ITAE retain their classical meaning as measures of tracking accuracy, accumulated regulation error, and persistence of late-time deviations. On the other hand, the indices J_u , $J_{\dot{u}}$, J_{e_u} , ITAE_{e_u} , and J_s are introduced to characterize the energetic and structural effects of fractionalization beyond conventional tracking assessment. In particular, J_u is associated with the overall actuation demand, $J_{\dot{u}}$ reflects control smoothness and high-frequency activity—hence relating to chattering propensity, actuator stress, and operational comfort—whereas J_{e_u} , ITAE_{e_u} , and J_s quantify the deviation of the fractional regime with respect to the integer-order reference case in terms of control action and sliding-manifold deformation. Therefore, the proposed set of indicators should be interpreted not as redundant metrics, but as a two-level evaluation framework linking tracking quality with energetic severity, smoothness, and structural control behavior (see Appendix 1).

6.7 Benchmark controller: classical PID

A classical continuous-time PID controller was established as a benchmark:

$$v_q(t) = K_p e_q(t) + K_i \int_0^t e_q(\tau) d\tau + K_d \dot{e}_q^{(f)}(t), \quad (62)$$

where $\dot{e}_q^{(f)}(t)$ represents a filtered derivative of the error.

The PID gains were adjusted to obtain a rapid and well-damped response:

$$K_p = 0.60, \quad K_i = 8.0, \quad K_d = 0.004. \quad (63)$$

This controller exemplifies a conventional industrial solution and acts as a significant reference point for assessing the advantages of fractional-order sliding mode control.

The fractional-order sliding mode control law is articulated as

$$v_q = L_q(-\ddot{\omega}_m^* + f_q(x) + k_1 I^\beta(\phi(s_f)) + k_2 s_f), \quad (64)$$

where

$$\phi(s) = \|s\|^{1/\beta} \text{sgn}(s), \quad 0 < \beta \leq 1. \quad (65)$$

The controller gains were established as follows:

$$k_1 = 20, \quad k_2 = 5, \quad \lambda_q = 15. \quad (66)$$

Fractional operators were realized through a fifth-order Oustaloup recursive approximation across the frequency range $[10^{-3}, 10^3]$ rad/s, which was discretized using the bilinear (Tustin) transformation.

6.7.1 Stage 1: PID versus fractional sliding mode control

The PID controller was evaluated in comparison to the fractional sliding mode controller, which was set at a fixed order of $\beta = 0.6$. All performance indices were evaluated to assess tracking accuracy, smoothness, and robustness.

6.7.2 Stage 2: fractional order sensitivity analysis

The second stage examined the impact of the fractional order

$$\beta \in \{0.1, 0.3, 0.6, 0.9, 1.0\}. \quad (67)$$

The case of integer order, specifically $\beta = 1$, served as a smooth nonlinear reference. The integration of RMSE, ITAE, IAE, J_u , $J_{\Delta u}$, and ITAE_s facilitated the identification of the fractional order that achieves the optimal balance among accuracy, robustness, and control smoothness.

7 Results and discussion

This section presents and examines the numerical results derived from the proposed FHGO-FOSMC scheme within the simulation framework outlined in the preceding section as shown in Figure 1. The assessment is structured into phases. Stage I establishes a direct comparison between a classical PID controller and the proposed fractional sliding-mode strategy (with a fixed value of $\beta = 0.6$), while maintaining identical plant conditions and utilizing the same fifth-order speed reference. Stage II (to be discussed subsequently) examines the sensitivity concerning the fractional order $\beta \in \{0.1, 0.3, 0.6, 0.9, 1.0\}$, using the integer-order case $\beta = 1$ as the structural reference. The discussion consistently emphasizes (i) the behavior of time-domain tracking, (ii) the accuracy of steady-state regulation, and (iii) the indices of aggregate error-energy, with a specific focus on the robustness in the face of nonlinear friction and speed-dependent load effects inherent in the plant model.

7.1 Stage I: PID versus fractional SMC ($\beta = 0.6$)

Figure 2 provides a summary of the Stage I comparison. In Figure 2a, both controllers adhere to the specified speed trajectory; nonetheless, the proposed fractional SMC demonstrates a noticeably tighter tracking performance during the acceleration ramps and speed transitions. The PID response demonstrates a persistent lag and significant deviations in response to reference changes, illustrating the trade-off between bandwidth and robustness that is characteristic of fixed-gain PID regulation in the context of nonlinear electromechanical coupling and frictional influences. In contrast, the fractional SMC trajectory closely aligns with the reference throughout the same intervals, demonstrating enhanced

closed-loop damping and disturbance rejection while avoiding overshoot.

The steady-state behavior is elucidated by the magnified perspective presented in Figure 2b. During the chosen time interval, the output of the PID controller exhibits a distinct offset in relation to the reference, while the fractional Sliding Mode Controller (SMC) maintains a considerably closer alignment with the setpoint. The residual bias holds significant importance in traction drives, as it directly indicates the ongoing speed regulation error resulting from the interplay of Coulomb friction, rolling losses, and aerodynamic torque components. The proposed controller demonstrates effective compensation for these effects, resulting in a steady-state deviation that is significantly reduced compared to the PID baseline.

Figure 2c provides additional evidence for these observations via the tracking error trajectories. The PID error experiences relatively significant fluctuations during transients and demonstrates an extended recovery period following reference changes, characterized by a notable negative deviation during the deceleration phase. This behavior aligns with a restricted resilience to nonlinear load and friction terms, as well as uncertainties that depend on parameters. Conversely, the fractional SMC error stays close to zero throughout the entire horizon, exhibiting only minor perturbations around the switching instants. This indicates a more rapid contraction of errors and enhanced attenuation of matched disturbances.

Ultimately, the worldwide significance of these enhancements is measured through the RMSE bar comparison illustrated in Figure 2d. The proposed fractional SMC achieves a significantly lower RMSE compared to the PID controller, effectively addressing both transient deviations and steady-state bias throughout the entire experiment. In summary, Stage I illustrates that under identical reference and nonlinear plant conditions, the introduced fractional SMC achieves enhanced tracking accuracy, diminished steady-state offset, and significantly better error-energy performance in comparison to the traditional PID baseline.

7.2 Stage I (continued): integral performance indices and control-effort assessment

In addition to immediate tracking plots, a thorough comparison necessitates aggregate indices that encompass the error dynamics and the control effort throughout the duration. Figure 3 illustrates the cumulative progression of the standard tracking criteria IAE and ITAE, alongside the control-energy index J_u and the instantaneous control signal $u_q(t)$ for both the PID baseline and the proposed fractional SMC with $\beta = 0.6$. This representation provides valuable insights for traction drives by distinguishing between *the rate of error accumulation* and *the extent of actuation* necessary to maintain the tracking behavior.

In Figure 3a, the cumulative IAE of the PID controller exhibits a rapid increase during the acceleration and transition intervals, persisting in its growth as further transients arise. Conversely, the fractional SMC curve stays near zero throughout the entire horizon, indicating that the proposed strategy effectively minimizes the absolute error in nearly all time intervals. This is not simply a temporary benefit: the ongoing distinction between the curves

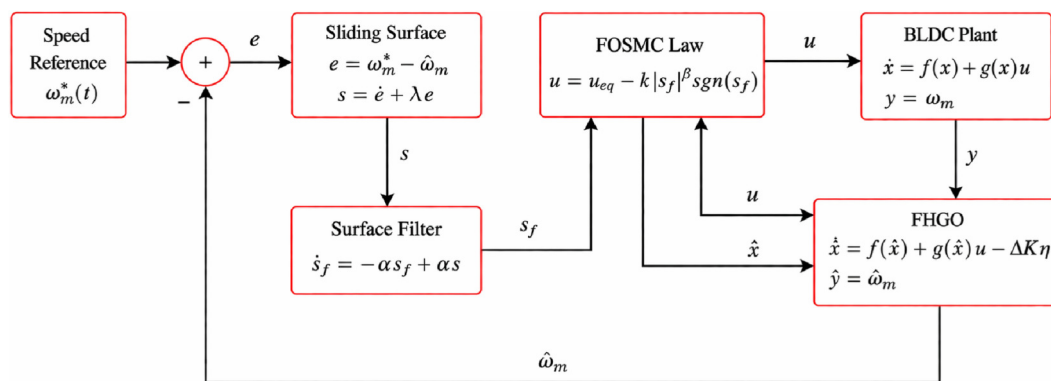


FIGURE 1

Proposed observer–controller architecture for the Brushless DC (BLDC) traction system. The diagram illustrates the closed-loop interconnection between the speed reference, the sliding-surface generation stage, the fractional-order sliding mode controller (FOSMC), the Brushless DC (BLDC) nonlinear plant, the surface filter, and the filtered high-gain observer (FHGO). The estimated speed $\hat{\omega}_m$ is fed back to construct the tracking error, while the filtered sliding variable s_f is used to regularize the control action. The shared information flow between the controller and the observer highlights the role of estimated states and control signals in the proposed robust output-feedback framework.

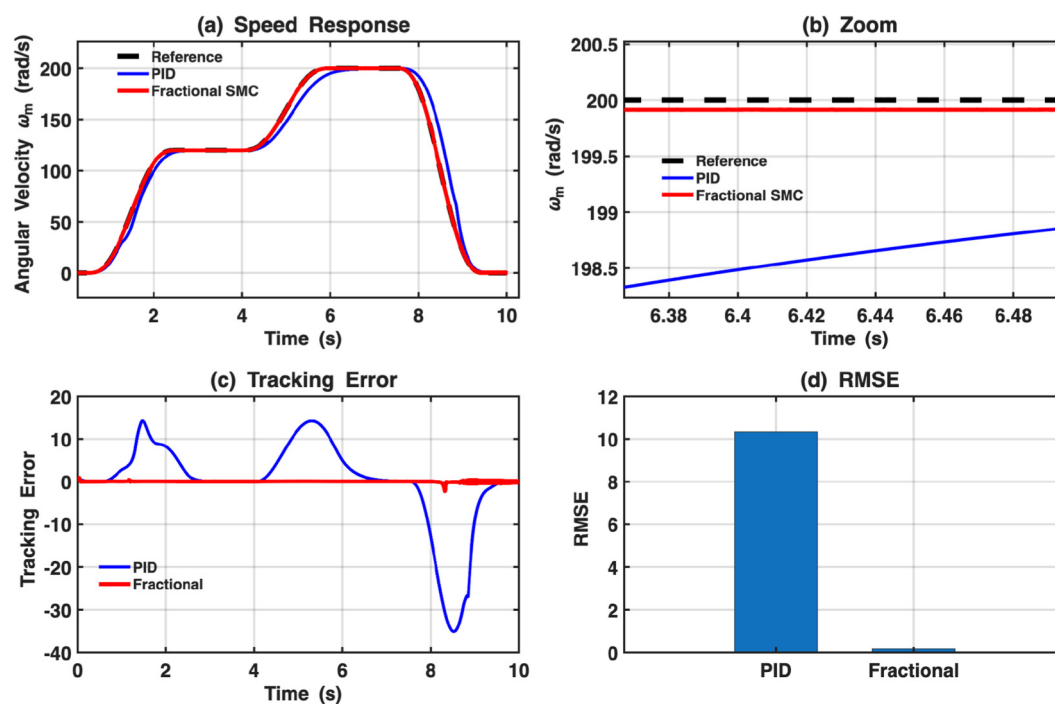


FIGURE 2

Stage I results: PID vs. proposed fractional SMC ($\beta = 0.6$) under identical plant conditions. (a) Comparison of speed response over the entire horizon, illustrating reference tracking. (b) A zoomed view (steady-state window) that emphasizes the fractional SMC's tighter setpoint adherence and the PID's regulation offset. (c) The proposed controller demonstrates considerably reduced transient excursions and faster recovery by tracking error trajectories. (d) A comparison of the RMSE over the entire horizon, which summarizes the global tracking accuracy enhancement achieved with fractional SMC.

suggests that the suggested closed-loop behavior consistently preserves a narrow error tube, even in the presence of nonlinear loads, Coulomb friction, and speed-dependent torque losses inherent in the system. Consequently, the enhancement is fundamentally structural rather than merely incidental, and is consistent with the anticipated robustness of sliding-mode-type dissipation mechanisms.

The trends observed in Figure 3b further substantiate this conclusion. Due to the nature of ITAE, it places a greater emphasis on late errors, resulting in a penalty for slow recovery and persistent steady-state bias. The PID demonstrates a significant increase in ITAE during the later stages of operation, suggesting that its regulation error is not only greater but also more enduring over time. In contrast, the fractional SMC produces an ITAE that is

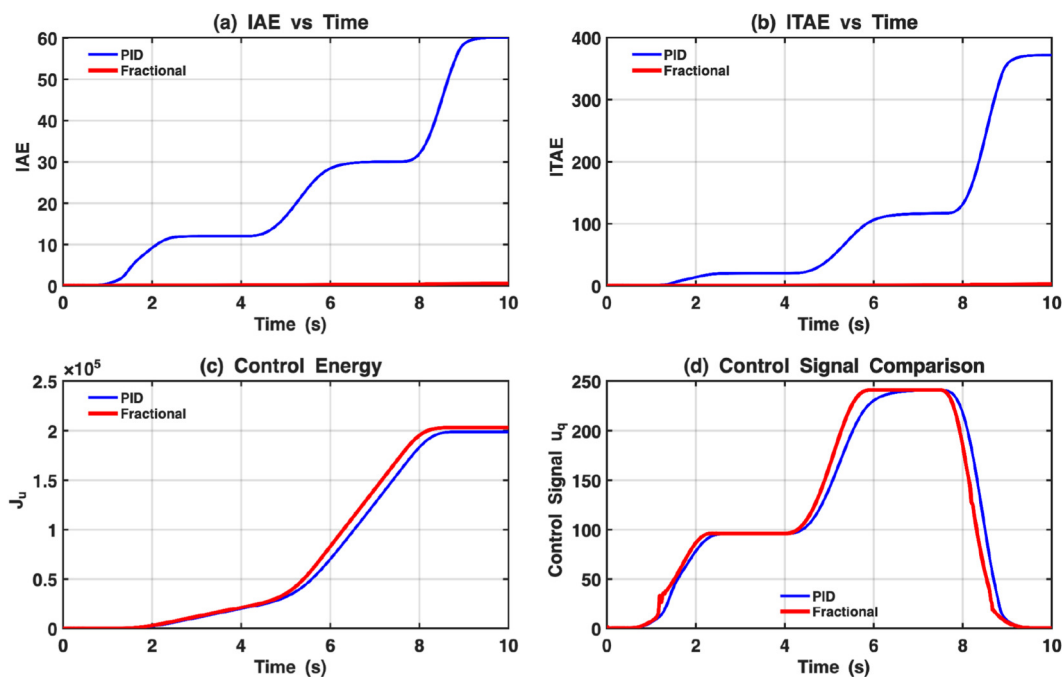


FIGURE 3

Stage I integral indices and control-effort comparison (PID vs. proposed fractional SMC, $\beta = 0.6$). (a) Time-integrated cumulative absolute error (IAE).

(b) Time-weighted absolute error cumulative integral (ITAE) versus time. (c) Cumulative control-energy index $J_u(t) = \int_0^t u_q^2(\tau) d\tau$. (d) The instantaneous control signal $u_q(t)$ for both controllers. The fractional SMC results in a significantly reduced accumulated error (IAE/ITAE) at the expense of a slightly higher control-energy and a mild residual disturbance near the end of the horizon.

notably minimal, indicating that the controller effectively promotes rapid convergence of the tracking error while mitigating prolonged deviations. From the perspective of electric vehicle dynamics, this characteristic is essential: ongoing speed discrepancies result in torque misallocation, reductions in efficiency, and compromised drivability in the presence of varying resistive torques.

Figure 3c illustrates the cumulative control-energy index defined as $J_u(t) = \int_0^t u_q^2(\tau) d\tau$. The fractional SMC demonstrates a marginally elevated final energy compared to PID. This aligns with a comprehensive approach that incorporates more vigorous measures to manage uncertainty and friction effects, especially in challenging transient conditions. It is noteworthy that the energy gap remains relatively modest when compared to the significant tracking gains observed in IAE/ITAE. The proposed method results in a substantial reduction in accumulated error, achieving orders-of-magnitude improvement, albeit with a minor increase in control energy. In practical terms, this trade-off is generally advantageous in traction applications, as enhanced speed regulation and diminished error persistence directly contribute to improved ride quality, consistent torque, and effective disturbance rejection. Furthermore, the slight increase in electrical actuation energy is frequently insignificant when compared to the total power flow within the drivetrain.

The instantaneous control trajectories depicted in Figure 3d elucidate the source of this discrepancy. The fractional control signal exhibits a somewhat more assertive behavior during transitions, which accounts for the elevated J_u . A subtle high-frequency ripple is observable towards the conclusion of the experiment, which may be interpreted as residual chattering resulting from the sliding-type

action interacting with nonlinear friction and the numerical filtering or approximation of fractional operators. Importantly, this ripple seems to be contained and does not affect tracking performance, suggesting that the proposed design successfully maintains robustness while avoiding destabilizing oscillations. From an implementation perspective, such residual chattering can be further mitigated by standard actuator-consistent measures (boundary-layer saturation, rate limiting, or refined fractional filtering bandwidth) without sacrificing the dominant robustness benefits. *Why PID serves as the baseline.* While there are more advanced industrial controllers available, PID continues to be the most commonly used solution in electromechanical and drive systems. This is attributed to its simplicity, low computational requirements, ease of tuning, and a robust history in industrial applications. As a result, benchmarking against PID offers a significant and cautious reference point: demonstrating clear superiority over PID not only establishes immediate practical value but also underscores the performance gap that drives the need for advanced nonlinear and fractional control methods. The results demonstrate that the fractional SMC significantly surpasses the PID baseline in terms of regulation quality and error persistence. This improvement occurs with only a slight increase in control energy and a minimal residual ripple, which can be addressed through standard implementation refinements.

It should be emphasized that the comparative analysis of this study is organized in two complementary layers. In Stage I, the proposed FHGO-FOSMC scheme is compared against a classical PID controller retained as the industrial benchmark, since PID remains the most widely used control structure in practical

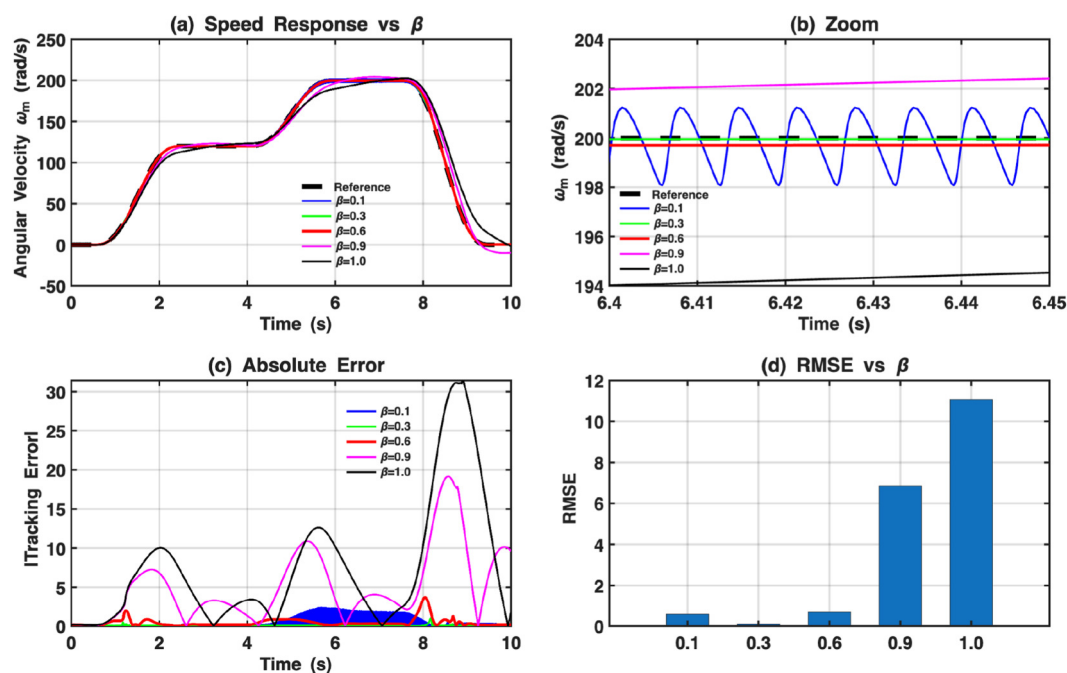


FIGURE 4

Stage II results: sensitivity analysis with respect to the fractional order β . (a) Comparison of the speed response for $\beta \in \{0.1, 0.3, 0.6, 0.9, 1.0\}$ under identical nonlinear plant conditions. (b) A zoomed steady-state window that emphasizes oscillatory behavior for small β and regulation bias for $\beta = 1.0$. (c) Absolute tracking error trajectories that demonstrate the smallest error envelope for intermediate β values. (d) The optimal performance region is indicated by the RMSE versus β , which is approximately 0.6.

electromechanical applications. Importantly, the PID benchmark is not driven by raw measurements, but by the same filtered high-gain observer (FHGO) adopted in the proposed architecture; therefore, both schemes share the same estimation layer, and the comparison more fairly isolates the effect of the control law itself. In Stage II, the analysis is extended through an intra-family structural comparison, where the integer-order case $\beta = 1$ is treated as a non-fractional observer-based nonlinear compensated reference configuration, while the cases $0 < \beta < 1$ represent the fractional regimes. This second comparative layer allows the specific contribution of fractionalization to be assessed independently of the common FHGO-based estimation structure.

It is important to emphasize that the absence of pronounced chattering in the proposed responses is a direct consequence of the controller architecture rather than a plotting artifact. In particular, the combination of the filtered sliding surface, the FHGO-based state estimation, and the fractional-order reaching dynamics regularizes the switching action and attenuates high-frequency oscillations in the control channel. This interpretation is further supported by the Stage II structural indices, especially the control-smoothness metric J_{ii} , which quantitatively shows that appropriately selected fractional orders achieve substantially lower high-frequency activity than excessively small β values.

7.3 Stage II: sensitivity analysis with respect to β

Stage II examines the proposed controller's sensitivity to $\beta \in \{0.1, 0.3, 0.6, 0.9, 1.0\}$ in order to further clarify the structural

influence of the fractional order on the closed-loop behavior. The current analysis isolates the internal dynamics of the fractional sliding mechanism itself, in contrast to Stage I, where the comparison was established against a classical PID baseline. The memory effect, which is introduced by the fractional operator, reshapes transient attenuation, steady-state regulation, and global tracking performance in nonlinear electromechanical conditions. This enables a direct assessment.

The primary dynamic responses are summarized in Figure 4. In Figure 4a, all fractional configurations are capable of adhering to the reference trajectory; however, their transient characteristics are significantly different. The response demonstrates increased oscillatory behavior and reduced attenuation at low β values, particularly $\beta = 0.1$ and $\beta = 0.3$, which are particularly evident during the acceleration and deceleration phases. The trajectory becomes smoother and more closely aligned with the reference as β increases toward intermediate values, particularly $\beta = 0.6$. This suggests an optimal compromise between high-frequency attenuation and integral memory action. The response gradually resembles a classical integer-order sliding formulation as β approaches unity, but its regulation performance is visibly degraded during rapid transitions.

The steady-state properties are further elucidated in the magnified window of Figure 4b. The structural role of β is more apparent in this context. For $\beta = 0.1$, a persistent oscillatory pattern appears around the setpoint, suggesting insufficient attenuation of matched disturbances and an overly reactive sliding manifold. This oscillatory amplitude is reduced as β increases; $\beta = 0.6$ maintains the closest adherence to the reference with minimal steady-state

disturbance. In contrast, the memory attenuation provided by the fractional operator is crucial for compensating the nonlinear friction and speed-dependent load torques embedded in the plant, as evidenced by the perceptible bias below the reference at $\beta = 1.0$. The findings indicate that the fractional order is not merely a tuning scalar, but a structural parameter that regulates the equilibrium between effective disturbance rejection and robustness.

These observations are further supported by the absolute tracking error trajectories depicted in Figure 4c. The configurations $\beta = 0.9$ and $\beta = 1.0$ exhibit substantially larger transient excursions, particularly during the final deceleration stage, where nonlinear friction effects predominate. Conversely, $\beta = 0.3$ and $\beta = 0.6$ result in significantly reduced peak errors and quicker contraction rates. It is important to note that the smallest overall envelope is achieved at $\beta = 0.6$, which suggests that the intermediate fractional order improves dissipativity while maintaining robustness against parameter-dependent uncertainties. Oscillatory residuals are introduced by extremely low β values, while high β values diminish the beneficial memory effect and impair regulation accuracy. It is important to note that the case $\beta = 1$ does not correspond to a classical discontinuous sliding-mode controller. Instead, it defines an integer-order observer-based nonlinear compensated regulation law acting on the filtered surface variable s_f . The model-based term $f_q(\dot{x})$ preserves the nonlinear compensation structure, while the integral contribution $\int_0^t s_f(\tau) d\tau$ replaces the fractional superlinear dissipation mechanism. Consequently, the $\beta = 1$ configuration serves as a meaningful non-fractional reference within the same observer-controller architecture, rather than as an external or unrelated benchmark.

In Figure 4d, the RMSE comparison quantifies the global impact of these behaviors. The bar chart plainly demonstrates that intermediate fractional orders, specifically $\beta = 0.3$ and $\beta = 0.6$, achieve the lowest RMSE values. Conversely, $\beta = 0.9$ and $\beta = 1.0$ demonstrate a significant decline in global tracking accuracy. This confirms that the fractional operator introduces a configurable memory depth that directly influences error accumulation across the entire horizon. In practical traction-drive applications, this results in reduced persistent regulation error and enhanced torque allocation precision when β is appropriately selected.

In general, Stage II illustrates that the fractional order β has a deterministic and structural impact on the efficacy of closed loops. The results suggest the existence of an optimal intermediate region (approximately $\beta \approx 0.6$) in which transient damping, steady-state accuracy, and global error metrics are simultaneously optimized. The performance is compromised by extremely small or unity values, either due to reduced disturbance compensation capability or oscillatory amplification. These results substantiate the importance of fractional-order tuning as a genuine design dimension, rather than a secondary gain adjustment. Stage II: Integral Performance and Control-Energy Analysis

The temporal evolution of the accumulated performance indices (IAE and ITAE), the control-energy functional, and the corresponding control signals for various fractional orders β is illustrated in Figure 5. The integer-order case ($\beta = 1$) and the intermediate fractional orders are clearly separated in panel (a), which corresponds to the cumulative IAE. The configurations $\beta = 0.3$ and $\beta = 0.6$ maintain substantially smaller slopes

throughout the entire horizon, despite the fact that $\beta = 1$ exhibits the largest accumulated growth. This behavior confirms that fractionalization modifies the dissipation mechanism in a structural rather than strictly parametric manner, as the fractional memory effect reduces the total absolute error accumulation not only during transient phases but also under steady-state conditions. This distinction is further emphasized by the ITAE curves in panel (b), which penalize persistent errors. The integer-order configuration exhibits accelerated growth beyond $t \approx 6$ s, which indicates a greater persistence of errors during the final operating phase. Conversely, the intermediate fractional orders exhibit growth rates that are substantially lower and under control. This confirms that fractional dynamics improve the rejection of nonlinear friction and speed-dependent load disturbances and enhance late-time error contraction. Despite the fact that $\beta = 0.1$ obtains a low accumulated error, its aggressive behavior is apparent in other metrics, which implies that oscillatory effects may be induced by excessively small fractional orders.

Panel (c) illustrates that performance enhancement is not energetically neutral. The configuration $\beta = 0.1$ necessitates the highest control energy, which is in accordance with the large-amplitude oscillations visible in panel (d). Residual chattering is evident. In contrast, $\beta = 0.6$ accomplishes an extraordinary compromise: it significantly reduces accumulated error indices while maintaining control energy that is comparable to the integer-order case. The actuation is smoother in the integer configuration, but the monitoring metrics are inferior. In conclusion, the structural role of β in shaping the dissipative geometry of the closed-loop vector field is validated by the existence of an optimal intermediate fractional region in Figure 5, which maximizes error reduction without excessive energetic penalty or oscillatory amplification.

7.4 Structural energy trade-off and optimal fractional order selection

The fractional sliding controller's structural energy behavior is synthesized in Figure 6 for a variety of exponent β values. The metrics illustrated in Figure 6 quantify the geometric deformation of the control law and the sliding manifold, in contrast to classical tracking indices. In particular, J_u quantifies the control curvature and chattering propensity, J_{e_u} assesses the energetic deviation from the integer-order case, $ITAE_u$ penalizes persistent structural discrepancies, and J_s depicts the deformation of the sliding surface dynamics.

The destabilizing effect of excessively small fractional orders is immediately apparent in the control smoothness index in Figure 6a. For $\beta = 0.1$, the value of $J_u(T)$ is 4.96×10^{10} , which is five orders of magnitude greater than the remaining configurations ($\sim 10^5$). This rapid expansion suggests that $\dot{u}$ exhibits severe high-frequency activity, which is indicative of near-discontinuous switching behavior. Chattering-like dynamics and diminished actuator longevity are directly linked to this energy amplification. In contrast, the smoothness levels of $\beta = 0.6$, $\beta = 0.9$, and $\beta = 1.0$ are comparable and bounded, which reinforces the notion that control curvature regularity is preserved by moderate fractionalization.

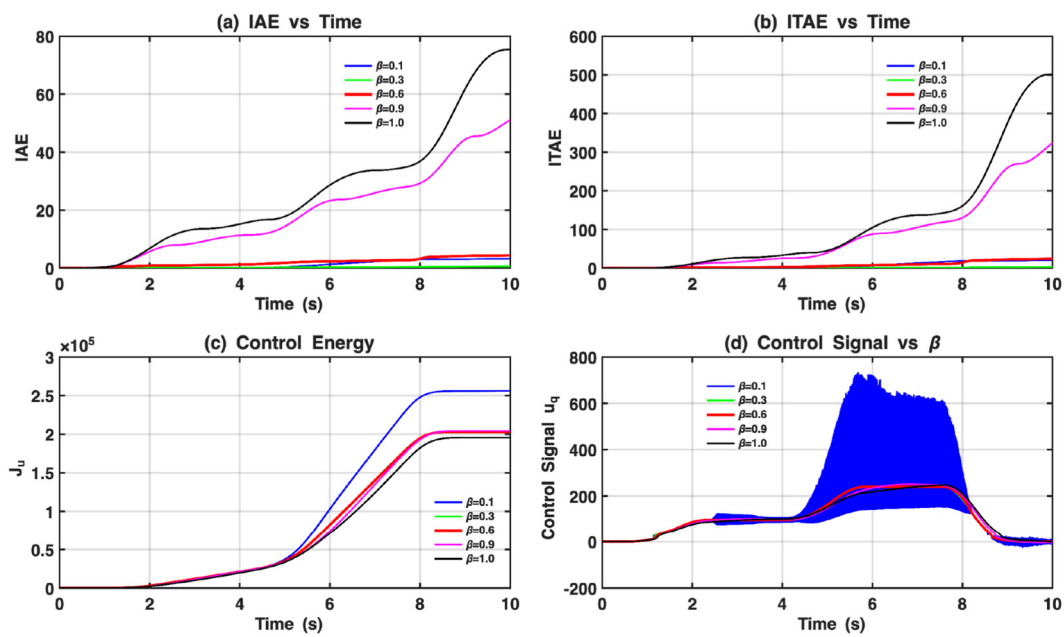


FIGURE 5

Stage II integral performance indices and control effort comparison for different fractional orders β . (a) Cumulative IAE versus time. (b) Cumulative ITAE versus time. (c) Control-energy functional $J_u(t)$. (d) Control signal comparison highlighting oscillatory behavior for low β .

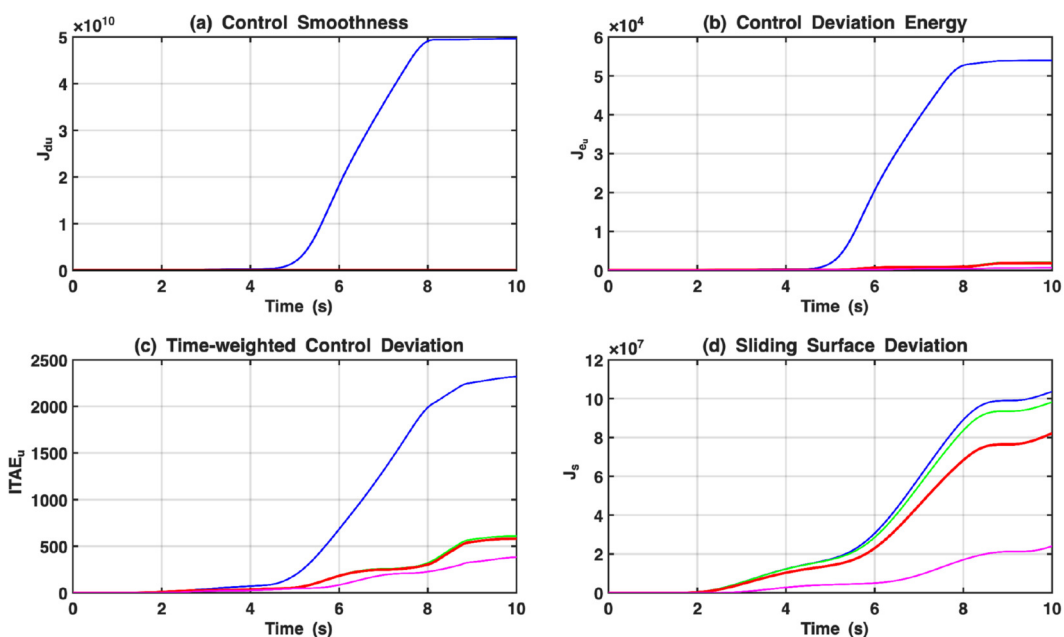


FIGURE 6

The fractional sliding controller's structural energy analysis under varying exponent values β . (a) The control smoothness index $J_{du} = \int_0^T (\dot{u}_\beta)^2 dt$ quantifies the propensity for chattering and oscillatory behavior. (b) The control deviation energy $J_{eu} = \int_0^T (u_{\beta=1} - u_\beta)^2 dt$ is a measure of structural deviation from the integer-order case. (c) The time-weighted control deviation $ITAE_u = \int_0^T t |u_{\beta=1} - u_\beta| dt$ is used to penalize persistent late-time discrepancies. (d) The geometric deformation of the sliding manifold is captured by the sliding surface deviation $J_s = \int_0^T (s_{\beta=1} - s_\beta)^2 dt$. The findings indicate that the optimal global trade-off between structural consistency, robustness, and smoothness is achieved at $\beta = 0.9$.

The control deviation energy J_{eu} shown in Figure 6b offers a glimpse into the extent to which the fractional controller deviates from the integer-order sliding regime. As β increases from 0.3 to 0.9, a

distinct monotonic reduction is observed, culminating at 6.43×10^2 at $\beta = 0.9$. This suggests that the structural properties of classical sliding mode control are gradually regained as the fractional exponents

increase, while the fractional damping characteristics are maintained. In contrast, the control geometry is profoundly altered by excessive fractionalization, as evidenced by the substantial energetic deviation (5.40×10^4) that is produced by $\beta = 0.1$.

This conclusion is further supported by the time-weighted deviation index $ITAE_w$, as illustrated in Figure 6c. The reduction of this metric from 6.09×10^2 at $\beta = 0.3$ to 3.83×10^2 at $\beta = 0.9$ is indicative of improved asymptotic structural alignment, as it penalizes late-time discrepancies. The persistent oscillatory behavior and long-term energy dispersion are reflected in the substantially larger value for $\beta = 0.1$ (2.32×10^3). Consequently, the structural disparity is diminished in both the transient and steady-state states as β increases.

In Figure 6d, the sliding surface deviation J_s is the most determining metric. As β approaches unity, a significant reduction is observed, with $\beta = 0.9$ achieving 2.41×10^7 , a value that is significantly lower than $\beta = 0.6$ and $\beta = 0.3$. Due to its substantially larger magnitude, the integer-order case $\beta = 1$ is not explicitly depicted, as it would cause the subplot scaling to be distorted. This fact is in itself enlightening: pure integer sliding generates a greater manifold energy than fractional configurations that are correctly tuned. Subsequently, the optimal compromise is $\beta = 0.9$, which maintains structural robustness and seamless control behavior while minimizing sliding manifold deformation.

In conclusion, the findings verify that β functions as a structural bifurcation parameter. Energy inflation and non-smooth actuation are induced by extremely small values, while values that are too close to unity restore classical behavior without fully utilizing the advantages of fractional damping. A balanced regime is represented by the range $\beta \in [0.6, 0.9]$; however, the optimal global trade-off among smoothness, energetic efficiency, robustness, and manifold stability is achieved at $\beta = 0.9$. Consequently, the optimal fractional exponent for the proposed controller is $\beta = 0.9$, as determined by the multi-index structural analysis of Figure 6.

8 Conclusion

This study presents a cohesive observer-controller framework for nonlinear Brushless DC (BLDC) traction systems, integrating filtered high-gain state estimation alongside a structurally adaptive fractional sliding mode controller. The study revealed that, in addition to traditional tracking enhancements, the fractional exponent β functions as a bifurcation parameter that regulates the geometric dissipation mechanism of the sliding manifold.

The theoretical analysis demonstrated exponential convergence under nominal conditions and input-to-state stability in the presence of bounded disturbances, with a clear focus on the interaction between the observer and controller, as well as the dynamics of filtering. In contrast to integer-order sliding control, which simplifies to linear integral regulation as β approaches 1, the fractional regime introduces nonlinear superlinear dissipation. This characteristic not only enhances robustness but also maintains smoothness when appropriately adjusted.

Comprehensive numerical analysis has uncovered a complex structural trade-off: excessively small β values lead to an increase in control curvature and result in high-frequency energy amplification, while the integer-order scenario demonstrates diminished robustness

and heightened manifold deformation. The intermediate region $\beta \in [0.6, 0.9]$ represents the optimal compromise, where $\beta = 0.9$ results in the minimum global structural energy while ensuring that the actuation effort remains within acceptable limits.

The results indicate that fractional sliding mode control should be regarded as a geometric energy-shaping approach, distinguishing it from being merely a parametric extension of traditional sliding mode control. The proposed framework presents a solution that is both mathematically rigorous and practically viable for next-generation electric vehicle drives, ensuring that robustness, smoothness, and structural efficiency coexist under stringent electromechanical constraints. Future research will focus on experimental validation using hardware-in-the-loop platforms and the extension towards adaptive fractional exponents that enable real-time structural modulation.

Although no real-motor experiments were included in the present study, the proposed FHGO-FOSMC architecture was intentionally developed first at the theoretical and nonlinear simulation levels in order to establish its structural and stability properties under realistic perturbations. Practical implementation is expected to involve additional challenges, including finite-order realization of fractional operators, sampling and quantization effects, observer tuning under noisy measurements, and inverter-induced actuation constraints, which motivate a dedicated hardware-validation stage as a natural continuation of this work.

Data availability statement

The original contributions presented in the study are included in the article/supplementary material, further inquiries can be directed to the corresponding author.

Author contributions

AR-M: Investigation, Software, Data curation, Conceptualization, Resources, Visualization, Funding acquisition, Writing – review and editing, Project administration, Formal Analysis, Writing – original draft, Methodology, Supervision, Validation. VG-H: Conceptualization, Investigation, Writing – review and editing, Validation, Formal Analysis, Methodology. GF-C: Conceptualization, Writing – review and editing, Methodology, Investigation, Formal Analysis, Writing – original draft. RL-P: Software, Writing – review and editing, Resources, Validation. PA: Investigation, Methodology, Writing – review and editing, Writing – original draft. OB-G: Data curation, Writing – review and editing, Software, Writing – original draft.

Funding

The author(s) declared that financial support was received for this work and/or its publication. This research was funded by Tecnológico Nacional de México (TecNM) under Grant No. CI-01/2026 (Project No. 24879), entitled “Plataforma automatizada de control, inteligencia artificial y observadores de estado avanzados

para la valorización de componentes bioasimilables mediante oxidación avanzada y fotobiorreactores”.

Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that generative AI was not used in the creation of this manuscript.

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Appendix A

Definition and Interpretation of Performance Indices

This appendix provides a formal definition of all performance indices used throughout the manuscript, together with their physical interpretation within the context of fractional-order sliding mode control applied to Brushless DC (BLDC) traction systems. The indices are organized into two categories: (i) classical tracking performance metrics, and (ii) structural and energy-based indices that characterize the effect of fractionalization on the control law and sliding manifold.

From a control-theoretic perspective, the first group of indices (RMSE, IAE, ITAE) evaluates tracking performance in a classical sense. In contrast, the second group (J_u , $J_{\dot{u}}$, J_{e_u} , $ITAE_u$, and J_s) provides a structural characterization of the control law, linking energy consumption, smoothness, and geometric deformation of the sliding manifold.

Therefore, the proposed evaluation framework should be interpreted as a dual-layer methodology: it not only measures tracking accuracy, but also reveals how fractional-order dynamics reshape the energetic and structural properties of the closed-loop system.